

In-situ pressure and temperature measurements under mixed friction in rolling contacts and comparison with TEHD simulations

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ABSTRACT

In-situ sensor systems can enable to measure the temperature and pressure of rolling contacts during mixed friction. A thin-film sensor is optimized with regard to its size and wear resistance. Traction tests are carried out with a two-disk test rig. The temperature and friction measurements are presented and compared with the results of a thermal elastohydrodynamic (TEHD) model. In a final step, the thin-film sensor is used in a cylindrical roller thrust bearing. Due to the detailed characterization the data are suitable for evaluating calculation methods. The comparison shows good agreement with the experimental data of a two-disk test rig. Further investigation is required in order to gain a deeper understanding of the processes that occur in real machine elements.

Nomenclature

a, b, c	-	Coefficient for fluid equation
A, B, C, D, E	-	Coefficient for fluid equation
a _s	m	Distance between pressure and temperature sensor
c _p , c _v	kJ/kgK	Specific heat capacity
D	m	Diameter
E _{Red}	N/m ²	Reduced Young's modulus
F _f	N	Friction force
F _N	N	Normal force
F _S	N	Sensor force
h _c , h _{min}	m	Central film thickness, minimum film thickness
L _f , L _{sol}	%	Fluid contact ratio, solid contact ratio
p _c	N/m ²	Solid contact pressure
p _{cav}	N/m ²	Cavitation pressure
p _{EHD}	N/m ²	EHD pressure distribution
p _h	N/m ²	Hydrodynamic pressure
p _{Hertz}	N/m ²	Hertz pressure distribution
p _{Hmax}	N/m ²	Maximum Hertzian pressure
SRR	%	Slide-to-roll ratio
T _{Bea}	°C	Bearing temperature
T _{Oil}	°C	Oil feed temperature
u _{sl}	m/s	Sliding velocity
u _{rol}	m/s	Rolling velocity
u ₁ , u ₂	m/s	Velocities in x direction
v	m/s	Velocity in y direction
w	m/s	Velocity in z direction
x, y, z	m	Cartesian coordinates

(continued on next column)

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x _{Hertz}	-	Cartesian coordinate normalized with the Hertzian contact width
α _p	m ² /N	Pressure coefficient
α _T	1/K	Temperature coefficient
β _{th}	1/K	Coefficient of volume expansion
γ	1/s	Shear rate
Δp	N/m ²	Pressure change
ΔR, R ₀	Ω	Resistance change, output resistance
Δθ	K	Temperature change
η	Pas	Dynamic viscosity
θ	-	Gap fill factor
θ	K	Temperature
λ	W/mK	Thermal conductivity
ρ	kg/m ³	Density
τ	N/m ²	Shear stress
τ _c	N/m ²	Critical shear stress
τ _f	N/m ²	Friction shear stress
Φ ^p	-	Pressure flow factor
Φ ^S	m	Shear flow factor
Ω	m ²	Calculation domain

1. Introduction

Experimentally confirmed calculation methods enhance confidence in design methods for rolling contact machine elements. Consequently, safety margins can be reduced and higher power densities of machine

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elements can be achieved. To ensure this, the models must consider numerous variables and undergo sufficient experimental validation. A crucial parameter is the friction-induced temperature increase in rolling contacts, which is used, for instance, in designing the scuffing load capacity of gears. For this reason, it has been necessary for many decades to measure the temperature within tribological contacts and to characterize the influencing variables. One of the challenges of concentrated rolling contacts is the high shear stress under mixed friction. Mechanically optimized thin-film sensors represent a promising solution. These sensors enable precise in-situ temperature measurement with high spatial resolution. However, the influence of the simultaneously occurring pressure in the contact area on the temperature measurement must be considered or compensated for. With the measurement results and the characterization of the influencing variables, a comparison with calculation methods can be achieved. This provides an important basis for further enhancements in the design and analysis of machine elements subjected to rolling and for positively shifting their performance limits. As early as the mid-1960s, Orcutt [1] investigated glass-steel contacts with a pressure and temperature sensor and used his results to compare them with the TEHD theory. He used manganin, a copper-manganese-nickel alloy with a very low temperature coefficient of resistance (TCR), as the pressure sensor material. Nevertheless, he found a temperature influence in his measurements and corrected this with the temperature measurement values. Platinum was used for the temperature sensor material, but he did not consider it necessary to correct the pressure error, as he could not determine any pressure dependency of his temperature sensor. Schouten [2] discussed various sensor materials and their resistance behavior with simultaneous temperature and pressure changes and which metals and alloys can be used. In particular, he discussed alloying elements in such a way that a temperature or pressure error is minimized. He then used sensors with a width of 10 μm and a layer thickness of 30 nm. Like Orcutt, Kannel et al. [3] also used platinum and manganin. They placed both sensors exactly on top of each other in a sandwich structure and carried out measurements under high-pressure conditions (>1 GPa). They suggested correcting the temperature error in the evaluation by calculation, but pointed out that this procedure can only be carried out afterwards and is therefore not advantageous for the experimental setup. They then developed a self-compensating bridge circuit to correct the pressure and temperature error online. Safa et al. [4,5] again choose the combination of manganin and titanium as a pressure and temperature sensor. In a previous study [4], they still assumed cooling effects, but then used a temperature correction. However, the procedure is not described in detail.

Köhler [6] used manganin and a chromium-nickel alloy to measure the pressure and titanium for the temperature measurement. He described the pressure dependence on the deformation of the conductor and the change in the electrical conductivity of the material due to the volume change. He discussed temperature and pressure calibration and proposed a system of equations for correcting the pressure and temperature error. Additionally, he transferred the thin-film technology to radial cylindrical roller bearings of the NU 1010 and NU 1014 series, presenting pressure and temperature curves. These investigations included the impact of bearing clearance on the pressure distribution and the influence of bearing heating on contact temperature.

Baumann [7,8] used a combination of titanium and manganin sensors and described a two-layer sensor in which both sensors are placed on top of each other, resulting in no pressure correction being necessary. Therefore, he used different layer thicknesses to equalize the pressure coefficients of manganin and titanium. The temperature error of his two-layer sensor is so small that he does not have to correct the pressure error in the temperature measurement. In a more detailed work [7], he described the use of cover layers to protect the sensors from wear and short circuits in the event of mixed friction. He saw the thermal insulation effect of the cover layer as a major disadvantage. Peeken et al. used their titanium-manganin sensors directly on the tooth flank and

described how the pressure and temperature measurements must be corrected [9]. Kagerer et al. [10,11] used platinum and manganin as sensor materials. They described the influence of pressure and temperature on the change in sensor resistance and suggested that the ratio of the pressure to the temperature coefficient should be large and vice versa. They did not perform a signal correction but specified a maximum error for their temperature measurements. The sensors were used up to a pressure of 1500 MPa, although a pressure of 1000 MPa was selected for most of the experiments. The cam-tappet contact is examined in more detail by Knauss [12]. He used manganin as a pressure sensor and titanium as a temperature sensor. He also used very thin cover layers to protect his sensors from wear and short-circuiting in the event of mixed friction. He explained the error in temperature measurement due to the thermal insulation effect of the wear protection layer, but he was unable to provide an accurate error estimate. Kühl and Holand [13,14] placed their titanium and chrome sensors next to each other on the tooth flank to record the contact values simultaneously. They described the pressure and temperature changes in general as a polynomial but linearized their characteristics for the technically relevant operating range and then, like Köhler [6], used a system of equations. In a more detailed work, Kühl also dealt with cover layers as wear protection and analyzed the thermal insulation effect using an example. Sperrfechter [15,16] used the already proven variant of titanium and manganin Knauss from for his tests on ceramic-ceramic and ceramic-steel contacts. At a pressure of 1000 MPa and a temperature change of 50 K, he specified a temperature error of 5.2 % for titanium and a pressure error of 2.5 % for manganin. To correct the pressure and temperature influence, he determined the time offset of both signals and corrected both measurements using an iterative approach. He tried to avoid mixed friction but used a cover layer due to the high mechanical stress and for insulation reasons. However, he also mentioned the chemical influence of manganin, as the material changes its resistance in the atmosphere (see [7]). According to his investigations, the thermal influence of the 0.1–0.2 μm silicon oxide cover layers used is negligible. However, he pointed out that for thicker layers (>1 μm), the thermal insulation effect must be considered again. He also carried out friction measurements and discussed the influence of thermal conductivity on the measured friction coefficient. Dauber [17] continued the work of Baumann and Sperrfechter and focused on the measurement of steel-ceramic contacts with pure rolling or minimal sliding. He attempted to answer the position assignment of the pressure and temperature signal to carry out the correction proposed by Baumann [18]. Mayer [19] investigated friction forces in EHD contacts with different lubricants and surface structures to improve the prediction of friction forces. He used platinum and manganin for detailed measurements in tribo-contacts. A temperature correction is not mentioned but is probable. He was able to achieve good results with smooth surfaces. Current investigations by Hofmann et al. [20] measure the temperature and pressure in ceramic-plastic and steel-ceramic contacts. They use platinum and chromium sensors on a ZrO_2 disk. For correction, Dauber's and a specially developed concept were used for signal assignment. In a former study, the authors [21] presented a thin-film sensor for temperature measurements but without a simultaneous correction of the pressure error. A slip free premeasurement was necessary to eliminate the influence of the pressure on the temperature signal. But with slip the pressure distribution changes slightly and in real machine elements these premeasurements are not always possible.

Summarizing, the relevant literature shows that if the rolling contact temperature is to be measured, a measurement error due to the pressure dependence of the sensor material used must be considered, and vice versa for the pressure measurements. Compensation methods are available through a specific measuring circuit [3], an adjustment of the pressure and temperature coefficient [8] or subsequent correction using signal adjustment and solving a system of equations [6]. With modern measurement data acquisition and analysis, this correction can also be carried out in real-time at the test rig.

In many of the studies presented, mixed friction was purposely

avoided due to the high mechanical stress and resulting destruction of the sensor. An option is to introduce a cover layer as wear protection which also protects against short circuits in the event of mixed friction when the sensor material touches the counter body. However, a major disadvantage is the thermal insulation effect, even for coatings $< 1 \mu\text{m}$, as the layers utilized like Al_2O_3 exhibit thermal conductivity that is several times lower than that of steel. Such a wear protection layer makes it difficult to interpret the sensor readings in terms of the surface temperature and therefore the lubrication gap temperature in machine elements. Cover layers have already been used in the past, particularly for wear protection. However, error estimation due to the thermal insulation effect has not been sufficiently possible to this point or was not studied [7,12,16,21].

In this study, a temperature sensor made of titanium and a pressure sensor made of either chromium or manganin were employed to determine temperature and pressure simultaneously in the rolling contact of a two-disk test rig. The two sensors were placed one behind the other on one of the two disks. The distance in the direction of movement ($< 1 \text{ mm}$) was measured optically, and the pressure and temperature signals were compared using the disk surface velocity. Subsequently, the pressure and temperature errors were compensated according to Köhler [6]. The tests were carried out under high load and with fluid and mixed friction conditions. However, this requires a wear protection layer that prevents the sensor from mechanical wear, short-circuiting, and oxidation. The thin layers used, especially the wear protection layer, change the tribological contact.

To gain a deeper insight into the acting friction and contact temperature, a numerical TEHD simulation was conducted. This was done in conjunction with a mechanical and thermal characterization of the sensor and the lubricant, which then was used for comparison of the temperature, pressure and friction measurements with TEHD calculations. In this process, real surfaces and insulating effects of the coatings were taken into account.

In a review, Kirchner et al. [22] highlighted the significance, potential, and challenges associated with integrating sensors into machine elements. One potential technology is the use of thin-film sensors on the race tracks of rolling bearings. Consequently, in the final phase of this study, a rolling bearing test rig was utilized to examine the rolling contact between a roller and a washer. The sensor was thus positioned on a washer of a cylindrical roller thrust bearing.

2. Experimental setup

Initially, a two-disk test rig was used for extensive pressure and

temperature measurements. Subsequently, a rolling bearing test rig was employed for component tests. Therefore, the thin-film sensor development is described and the sensor materials, disks and bearing materials are presented focusing on their thermal and mechanical properties. At last, the test set ups and testing procedures are explained.

2.1. Test specimen

2.1.1. Pressure and temperature sensor

The structure of the thin-film sensor is depicted in Fig. 1 – left. Consisting of an insulation layer, two different structured sensor layers, and a wear protection layer, the coatings were applied to cylindrical steel disks (DIN 16MnCr5, AISI 5115 - two-disk test rig) and bearing washers (DIN 100Cr6, AISI 52100 - cylindrical roller thrust bearing). The insulating layer serves to prevent a short circuit with the steel base material. It is made of an aluminum oxide (Al_2O_3) layer deposited by physical vapor deposition (PVD). Photolithography combined with a lift-off process was used to structure the pressure and temperature sensor, which were coated in PVD processes. Optimization of the structuring process led to an improved spatial resolution compared to previous work [21] with a sensor width of $5 \mu\text{m}$, a length of $200 \mu\text{m}$ and a distance a_s of $800 \mu\text{m}$ between both sensors. The wear protection layer serves the purpose of avoiding material removal on the sensor layer during mixed friction and prevents a short circuit with the counter steel disk or rolling elements. A diamond-like carbon (DLC) layer, deposited in a plasma-enhanced chemical vapor deposition process (PECVD), was chosen for this. In preliminary tests, both the adhesion of the thin-film system as well as the wear resistance of the top layer were optimized for the use under mixed friction conditions and high loads. Fig. 1 - right illustrates a cross-section of an embedded temperature sensor using a scanning electron microscope. An ion etching process was employed to expose the cross-section, revealing the sensor positioned a few micrometers beneath the surface.

The measuring principle of the pressure and temperature sensor is based on the change in resistance due to mechanical pressure stress and temperature change. The total resistance change can be taken from Eq. (1).

$$\frac{\Delta R(t)}{R_0} \approx \alpha_T \Delta \theta(t) + \alpha_P \Delta p(t) \quad (1)$$

The sensors' pressure respectively temperature sensitivity can be described by the coefficients α_P and α_T . For the pressure measurement a large α_P and a small α_T is of advantage, for the temperature measurement vice versa. The coefficients are influenced by the material, layer

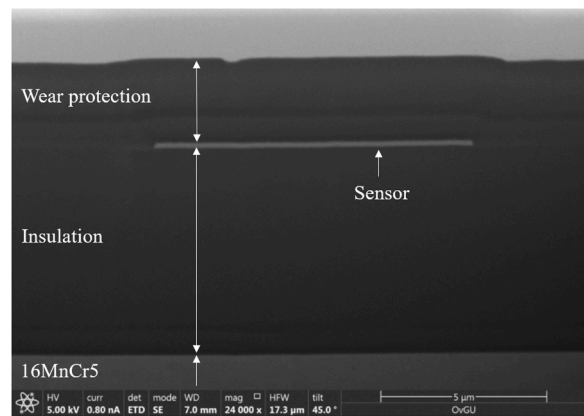
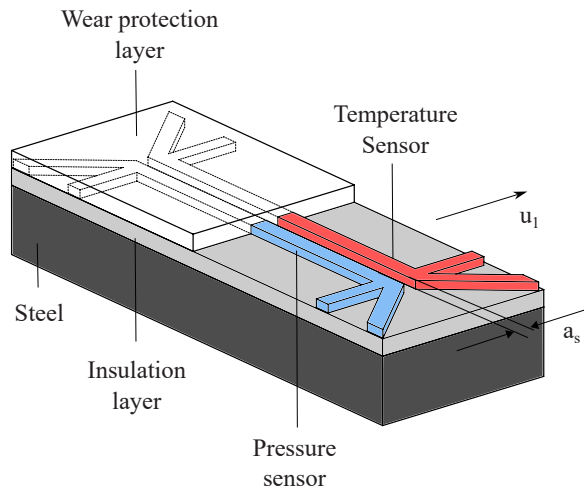


Fig. 1. Thin film pressure and temperature sensor – left; Electron microscope image of an embedded sensor as a cross-section (example) – right.

thickness, material purity and defects [2,6,16]. In particular, these influences make it necessary to calibrate each sensor [11]. Temperature coefficients were determined in a laboratory oven while pressure coefficients for the disks were determined in the two-disk test rig and with a laboratory press for the washers (Table 1).

The quantities pressure and temperature superpose in rolling contacts, which is why a signal correction is necessary [6]. It was purposely avoided to place both sensors directly on top of each other as Baumann reported [8], since this would have required additional layers and would have reduced the mechanical stability. Therefore, the two measured signals must first be aligned in the time domain. This can be done by knowing the distance a_s between the pressure and temperature sensor and the speed of the sensor disk u_1 (Fig. 1 – left). Subsequently, the pressure and temperature coefficients can be used to solve the equations proposed in [6,9,13,16,17]. Modern measurement equipment and software can perform this signal correction almost online and in real-time at the test rig.

Both chromium (disks) and manganin (washer) were used for the pressure sensor. On the one hand, manganin has a very low temperature coefficient and hence almost no thermal error on the pressure signal. On the other hand, chromium has a much higher pressure coefficient, allowing higher signal resolution in the measurements while the temperature influence can be corrected by the aforementioned calculations. For the temperature sensor, titanium was chosen due to its high temperature coefficient and relatively low pressure coefficient.

The pressure measurement is based on the piezoresistive effect and the change in conductor cross-section. The impact of frictional shear stress on the change in measured values remains difficult to determine. Therefore, before testing, sensors were applied to simple flat samples and subjected to shear stress. A minor impact on the pressure measurement was identified, with an estimated coefficient of friction of 0.07, which resulted in a maximum deviation of 1.78 % of the measured pressure.

2.1.2. Thermal and geometrical properties

The thermal and geometrical properties of the disks, washers and rollers play a significant role in determining the tribological contact conditions. In order to utilise these properties to set up a TEHD model and to compare the experimental temperature and pressure data with TEHD simulation data, the solid bodies and the sensor layers were extensively characterised in advance. The thermal insulation effect of the wear protection layer can be considered with the calculation methods used (Section 3). However, this requires the thermal properties and their temperature dependency of the disks and the sensor layers to be known.

As Beilicke et al. [23] have shown by simulation, a significant part of the frictional heat in a steel-DLC combination is conducted into the uncoated body due to the poor thermal conductivity of the DLC coating. Habchi et al. [24] studied the influence of soft annealed and hardened steel and emphasized the importance of the correct thermal conductivity. This follows the work of Reddyhoff et al. [25]. To be able to calculate the exact heat distribution, the thermal conductivities of the steel disks made of 16MnCr5 and washers made of 100Cr6 were first determined using the laser flash method over a temperature range from 30 °C to 250 °C. For this purpose, samples of case hardened 16MnCr5 (HRC 62), unhardened 16MnCr5, hardened 100Cr6 (HRC 62) and unhardened 100Cr6 were manufactured and measured. In the laser flash method

used, the specimen is exposed to a high-intensity laser radiation energy pulse of short duration on the top surface of the specimen. The energy is absorbed by the sample and re-emitted on the underside of the sample. This radiation causes a temperature rise on that side and is recorded by an infrared detector. With the knowledge of the density and the heat capacity, the thermal conductivity can be determined. Fig. 2 shows the dependence of thermal conductivity on temperature. It can be seen that the thermal conductivity is strongly influenced by hardening and both steels exhibit a slight decrease in thermal conductivity with increasing temperature.

The difference in thermal conductivity can be explained by the microstructural change of the steel. The unhardened 16MnCr5 has a (ferritic-)pearlitic and the hardened 16MnCr5 has a fine needle like structure. The unhardened 100Cr6 has pearlitic structure with small carbides and the hardened 100Cr6 has a very fine, needle-like matrix structure (Fig. 3).

The thermal conductivities of the layers of the thin-film system were determined as a function of the temperature using the 3-omega method. To this end, a sensor structure analogous to that used on the disks was employed, with one component serving as a heating element and the other as a sensor. By subjecting the system to periodic heating, the thermal conductivity of the Al_2O_3 insulation layer and the DLC wear protection layer could be derived from the sensor response. The thermal conductivity increases with higher temperature. It is assumed that this effect is linear within a temperature range of 20 – 180 °C (Fig. 4).

The layer thicknesses were measured by a profilometer on reference samples which were carried along during the preparation of the sensor layers. The density of the coatings was determined on silicon wafers and the specific heat capacity was taken from the literature (Table 2). The metallic sensor layers exhibit a significantly higher thermal conductivity than the insulation and wear layers, and are also notably thinner. Consequently, these layers were not considered in the simulation and were not characterised accordingly.

Similar to the analysis of thermal properties, both the macroscopic and microscopic geometry were considered. In Table 3, a comprehensive summary of significant geometric parameters influencing the tribological contact in the two-disk test rig and the cylindrical roller thrust bearing is provided. The diameter and surface microgeometry of the disks and rollers were determined as in [21]. It is essential that the surface of the sensor be of a mirror-like surface to prevent a short circuit resulting from the presence of roughness in the elevation between the substrate material of the disk respectively the washer and the sensor. Hence, the sensor disk and the sensor washer were subjected to vibro-polishing. In comparison, the counter disk and rollers are designed rougher. With these surfaces mixed friction conditions (as described in Section 3) could be achieved at speeds relevant to technical applications.

The creation of micro-contacts, and thus the real contact area, is significantly influenced by the surface structure (Table 3) and the plastic and elastic deformation of the materials. Consequently, the elastic-plastic behavior of the surfaces was measured (see Table 4). At the micro-contact level, contact forces generate high stresses over small areas, leading to material deformation. As a result, this plastic deformation increases the real contact area. To account for this in the simulation part of the study, the indentation modulus E_{IT} and Martens hardness HM were determined as described in [21].

2.2. Two-disk test rig

2.2.1. Set up

A two-disk test rig, previously described by the authors [21], was used to perform the measurements of pressure distribution and temperature rise in a rolling contact (Fig. 5). The pressure and temperature sensors are applied to the surface of a cylindrical sensor disk.

The rotation of the disks, along with the resulting surface velocities u_1 and u_2 drags heated oil into the gap, forming a supporting lubricating film. The half sum of both surface velocities, the so-called rolling

Table 1
Average pressure and temperature coefficients.

Sensor material	Pressure coefficient α_p (1/MPa)	Temperature coefficient α_T (1/K)
Manganin	$2.46 \cdot 10^{-5}$	$-0.06 \cdot 10^{-3}$
Chromium	$-8.01 \cdot 10^{-5}$	$0.75 \cdot 10^{-3}$
Titanium	$-0.94 \cdot 10^{-5}$	$2.21 \cdot 10^{-3}$

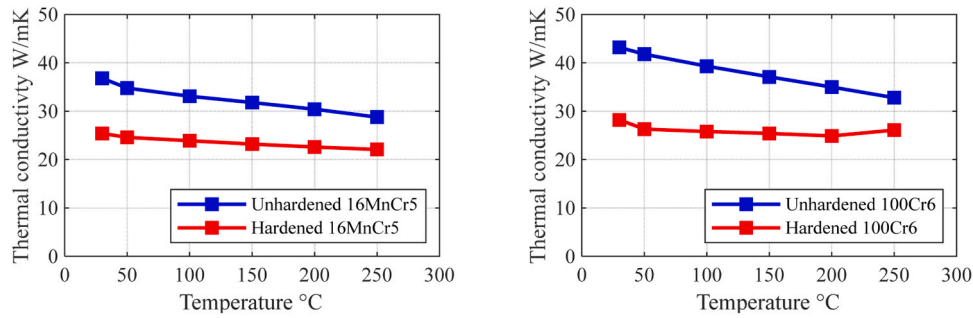


Fig. 2. Thermal conductivity of 16MnCr5 and 100Cr6.

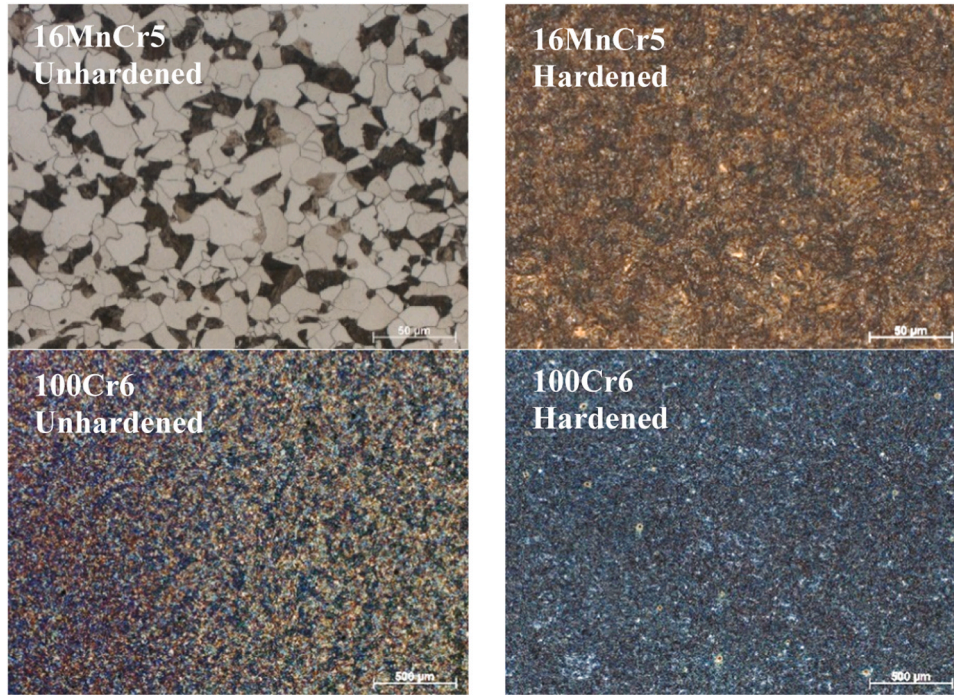


Fig. 3. Microscope images of microstructural change of unhardened and hardened steel.

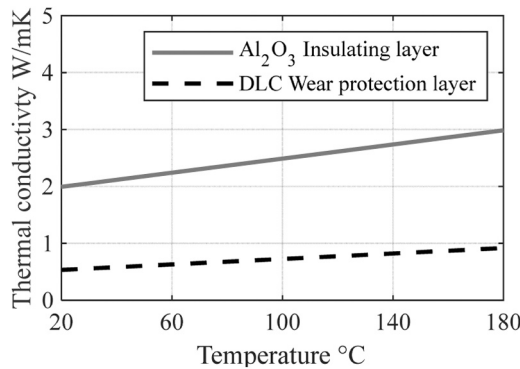


Fig. 4. Thermal conductivity of insulating and wear protection coating.

velocity u_{Rol} , is calculated as follows:

$$u_{\text{Rol}} = \frac{u_1 + u_2}{2} \quad (2)$$

The difference between the two tangential velocities is called sliding velocity u_{Sl} :

$$u_{\text{Sl}} = u_1 - u_2 \quad (3)$$

Table 2

Thermal properties of the thin-film sensor.

Coating	Material	Thickness (μm)	Thermal capacity (J/kgK)	Density (kg/m ³)
Insulation	Al ₂ O ₃	4.5	850	2900
Wear protection	DLC	2.5 (disk), 2.6 (washer)	700	2000

The slide-to-roll ratio SRR as the ratio of sliding velocity to the rolling velocity can be calculated as follows:

$$\text{SRR} = \frac{u_{\text{Sl}}}{u_{\text{Rol}}} = 2 \frac{u_1 - u_2}{u_1 + u_2} \quad (4)$$

The sliding of two surfaces in contact generates friction, which in turn produces heat. This in turn leads to an increase in contact temperature. The lubrication gap height can be systematically adjusted by controlling the oil feed temperature T_{Oil} and the rolling velocity u_{Rol} . In the presence of minimal lubrication gap heights within the level of the surface roughness, the probability of micro-contacts and, consequently, mixed friction increases. The friction force F_f resulting from the contact is measured with a tipping scale using a simple lever arm relationship. For this, one of the tables is rotatably mounted on a bearing block and the force sensor F_s is located between the bearing and the disk (Fig. 5).

Table 3
Macro- and micro geometry.

Experimental set up	Component	Macro geometry			Micro geometry (300 μm x 300 μm)		
		D (mm)	D _{Crown} (mm)	Finishing	Sa (μm)	Sq (μm)	Sz (μm)
Two-disk test rig	Sensor disk	65.000	-	Polished	0.019	0.025	0.298
	Counter disk	64.960	200.04	Grinded	0.201	0.259	2.167
Cylindrical roller thrust bearing	Washer	∞	∞	Polished	0.007	0.011	0.589
	Roller	10.997	∞	Grinded	0.176	0.231	2.090

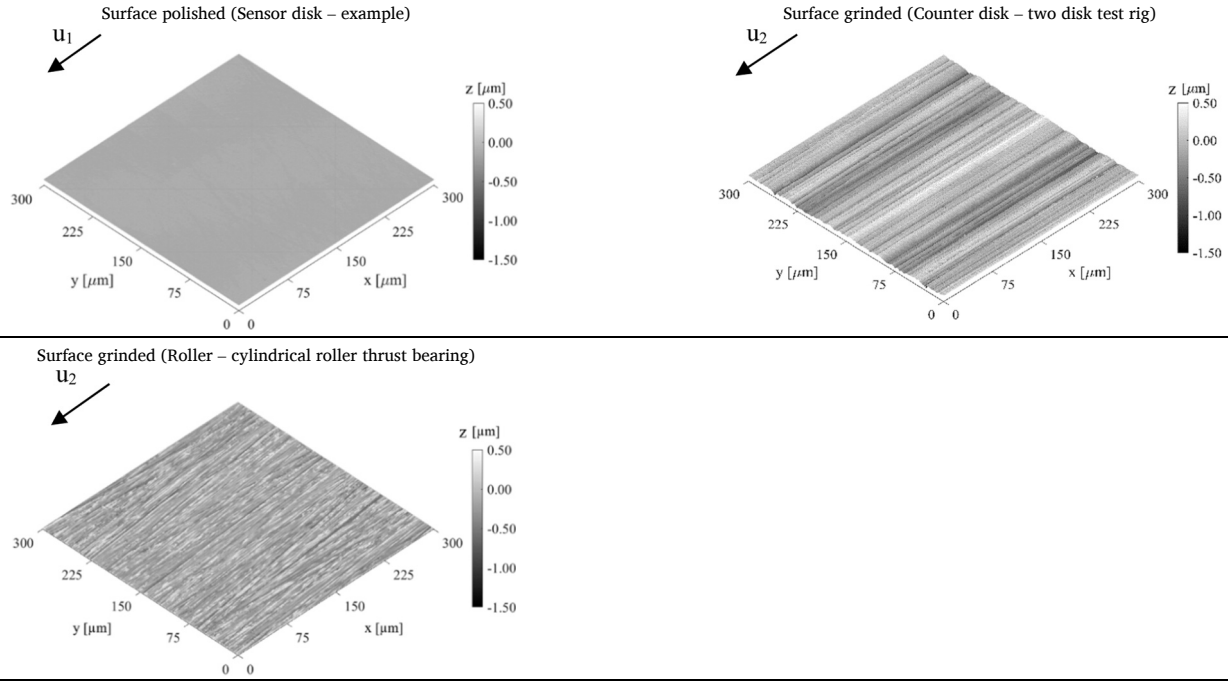


Table 4
Elastic-plastic properties of the surfaces.

Material (test force)	Indentation modulus E_{IT}	Martens hardness HM
DLC-layer (10 mN)	159 GPa $\pm$ 6 GPa	8384 MPa $\pm$ 144 MPa
Counter disk 16MnCr5 (1000 mN)	300 GPa $\pm$ 10 GPa	7500 MPa $\pm$ 210 MPa
Roller 100Cr6 (1000 mN)	334 GPa $\pm$ 13 GPa	8340 MPa $\pm$ 332 MPa

2.2.2. Test procedure

Shearing of a lubricant as well as sliding of surface-near boundary layers lead to a measurable integral friction and at the same time to a heat input in the contact area. To investigate simultaneously the shear behavior and the temperature rise under different operating conditions, so-called traction measurements were carried out. First, the two disks were accelerated to the rolling velocity u_{Rol} with the same speed each and then loaded with the normal force F_N . Subsequently, the friction force was measured continuously with increasing SRR but with otherwise equal conditions. The load-carrying capacity of the lubricant film is significantly influenced in the inlet area. If, for example, the bulk temperature of the disks increases due to the frictional heating, the oil viscosity and thus the lubricant film height will drop in the inlet area. To ensure an almost constant bulk temperature, the slip was increased in stages and after each stage pure rolling was performed for 20 seconds to cool down the disks. This is in compromise with the resulting test durations.

The initial tests were conducted to investigate the positive and negative SRR. However, due to the limited deviation in friction and contact temperature, the investigation was focused on the positive SRR. Various operating conditions were tested by adjusting the normal force, oil feed temperature, and rolling velocity. To generate mixed friction, several traction curves were run by decreasing the rolling velocity and

thus gradually reducing the lubricant film thickness (Table 5).

2.3. FE8 rolling bearing test rig

2.3.1. Set up

The FE8 rolling bearing test rig (Fig. 6 – left) in accordance with DIN 51819-1 is usually employed to carry out performance tests on lubricants [27]. For testing lubricating oils, the cylindrical roller thrust bearing of series 81212, made of hardened 100Cr6, is mainly used and standard tests in accordance with DIN 51819-3 or in-house standards are conducted with mixed friction. This bearing type was chosen due to its simple geometry and kinematics compared with others. The planar washers allow the application of the thin-film sensors with manageable effort concerning surface finishing and the structuring process. Cylindrical roller thrust bearings consist of two bearing washers and a cage with cylindrical rollers between the washers. In the FE8 test rig two test bearings are clamped with a spring assembly to apply the normal force F_N and the rolling elements are guided by the cage. The bearings are supplied with lubricant via two oil supply holes. The bearing temperature T_{Bea} is measured at the housing washer edge and controlled by heating and cooling of the housing. The resulting friction torque is measured via a force sensor mounted on the housing.

Due to their design, cylindrical roller thrust bearings exhibit

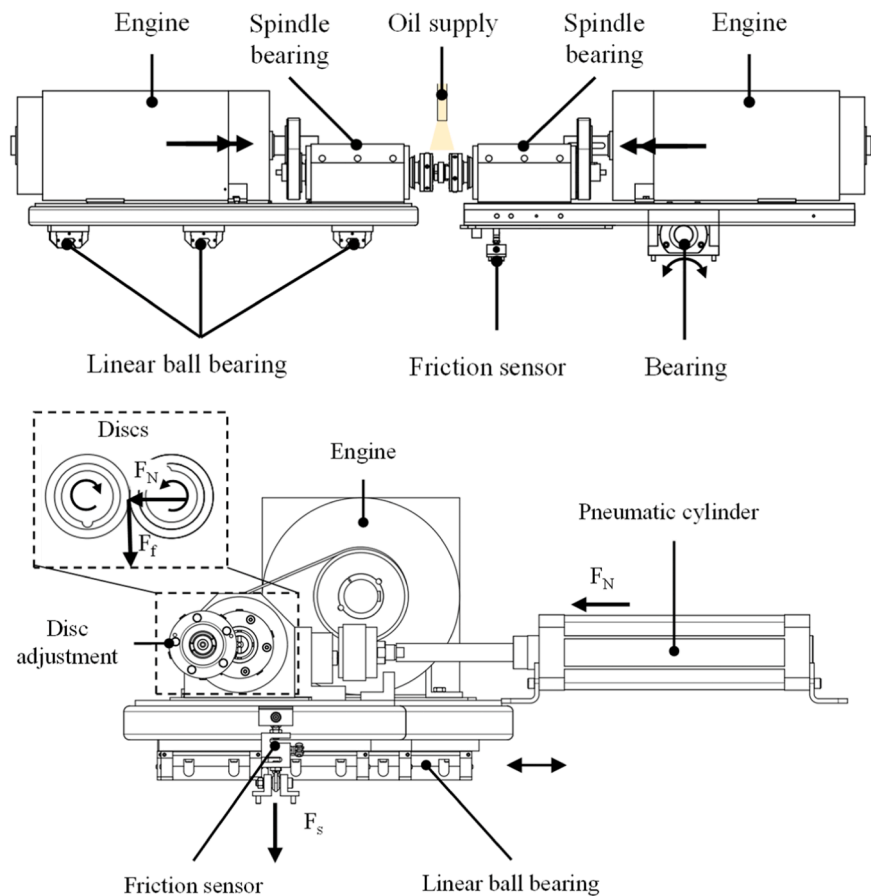


Fig. 5. Two-disk test rig [21].

Table 5
Test conditions with the two-disk test rig.

Parameter	Variable	Value
Oil feed temperature	T_{Oil}	40 °C, 60 °C
Normal force	F_N	400 N - 2000 N
Maximum Hertzian pressure	p_{Hmax}	1009 MPa - 1600 MPa
Rolling velocity	u_{Rol}	1 - 4 m/s
Slide-to-roll ratio	SRR	0 - 35 %
Lubricant	Reference Oil FVA 3 (ISO VG 100 mineral base oil, paraffinic basic solvent raffinates) [26]	

kinematic slip. The slide-to-roll ratio changes along the rolling element raceway. In the center of the raceway, there is almost pure rolling between the rolling elements and the bearing washers. In the inner and outer areas, SRR can decrease respectively increase by approximately - 9.8–8.5 %. The exact slip values depend on the rolling element geometry, the contact length, and the cage pockets. They were determined in advance to position the thin-film sensor in a manner that enables the detection of the maximum temperature at a given normal force. Therefore, the raceway on the bearing disc was initially analyzed to determine the area of pure rolling and maximum slip. Subsequently, the pressure and temperature distributions were calculated for several loads using the TEHD model described in chapter 3. Fig. 7 illustrates three

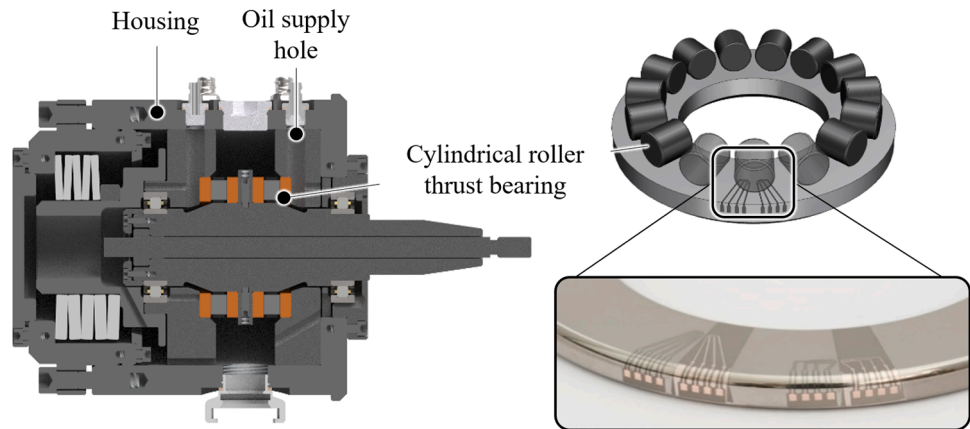


Fig. 6. FE8 test head – left; thin-film sensor on washer – right.

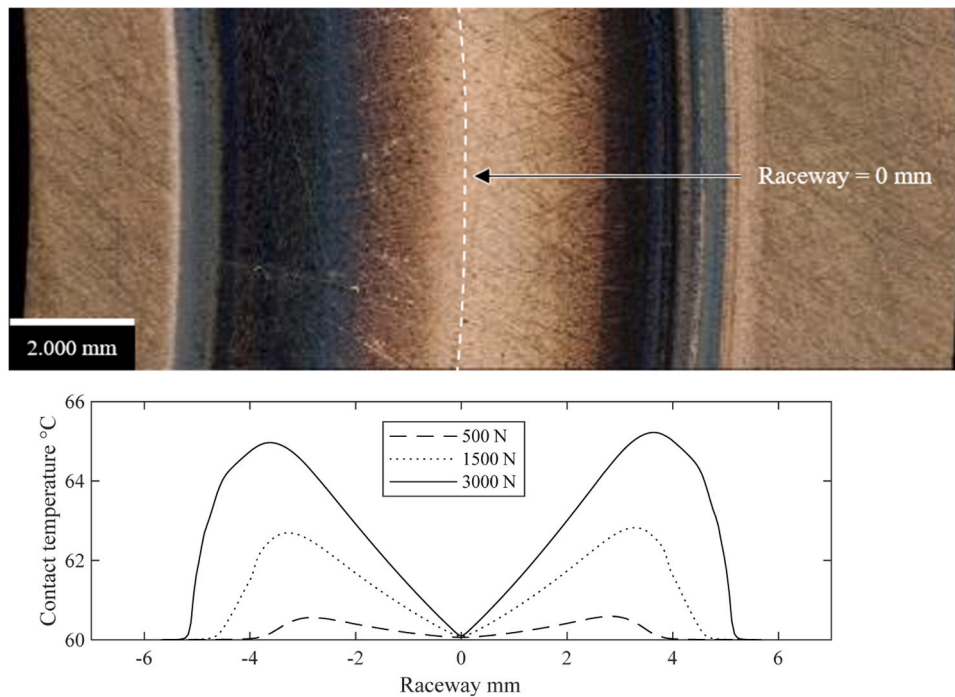


Fig. 7. Washer with raceway – above; example of the calculated contact temperature along the raceway for different roller loads at $n = 750$ rpm, $T_{\text{Bea}} = 60$ °C – below.

distinct temperature distributions along the raceway. As the applied friction power increases due to increasing slip towards the outside, the temperature rises. When a higher load is present, both the temperature increases due to higher frictional power respectively heat and the contact area increases due to the elastic deflection. Using this data, six different sensor positions were set up for the ability to investigate the load cases from Fig. 7 and the effect of slip in the inner and outer raceways. For the present study, only one sensor at a raceway position of 2.86 mm was used. The thin-film sensor is applied to the housing washer (Fig. 6 – right). The outer edge was prepared with a larger radius so that structuring of the conducting paths around the edge was feasible where wiring of the sensors was done. The test head was adapted accordingly to allow the wire routing from the washer to the constant current source and data acquisition box respectively. In order to compare the experimental data with the simulation, it was necessary to determine the exact position of the applied sensor on the raceway. This was achieved by scanning the surface with a microscope.

2.3.2. Test procedure

To allow further investigation with the sensory bearing, mixed friction was not applied at this stage to avoid the risk of wear and the load F_N was kept at 20 kN respectively at 884 MPa. The test rig was heated to the bearing temperature, and then the speed was increased to 1000 rpm and maintained for several minutes before the measurements were carried out. The test conditions are shown in Table 6.

Table 6

Test conditions for fluid friction with the FE8 test rig.

Parameter	Variable	Value
Bearing temperature	T_{Bea}	60 °C
Normal force	F_N	20 kN
Maximum Hertzian pressure	P_{Hmax}	884 MPa
Rotation speed	n	1000 rpm
Cage	Polyamide cage with 19 rollers	
Lubricant	Reference Oil FVA 3 A (ISO VG 100 mineral base oil, paraffinic basic solvent raffinates with 4 % EP-Additiv) [26]	

3. TEHD-simulation

The processes occurring within rolling contacts are complex and significant efforts have been dedicated over several years to comprehensively describe them through analytical, numerical, and experimental methods. The aim has been to identify potential optimizations for various technical applications [28–32]. In particular, the lubrication gap height is the focus of investigation, as it allows fundamental statements to be made about the wear and friction of the system. The reduction of wear and friction in lubricated tribological systems is typically accomplished by employing hydrodynamic lubrication to separate the contact surfaces. This method involves the formation of a lubricating film enabled by the movement of the lubricant into the decreasing gap. The extension of this concept to elastic contact partners, describing the flattening of the contact area, is known as elastohydrodynamic lubrication (EHD). The additional consideration of thermal effects leads to thermal elastohydrodynamic lubrication (TEHD). Fig. 8 shows the typical TEHD pressure, gap height and temperature distribution in the cross-section of the contact in the direction of movement, exhibiting the characteristic pressure and temperature spike at the end of the contact [32]. The lubrication gap can be approximately characterized by the central film thickness h_c or the minimum film thickness h_{min} . The TEHD simulation model used in this study is presented briefly in the following sections. A more detailed description of the model can be found in [23,33–35]. The simulation model has since been transferred to the commercial software Tribo-X, which was used for the calculations [36].

3.1. Elastohydrodynamic

The TEHD model is based on the numerical solution of the generalized Reynolds equation [37] for the three-dimensional contact under consideration of mass conserving cavitation by introducing the gap fill factor θ according to [38] and under the assumption of a solely movement in x direction, which aligns with the use cases in this paper:

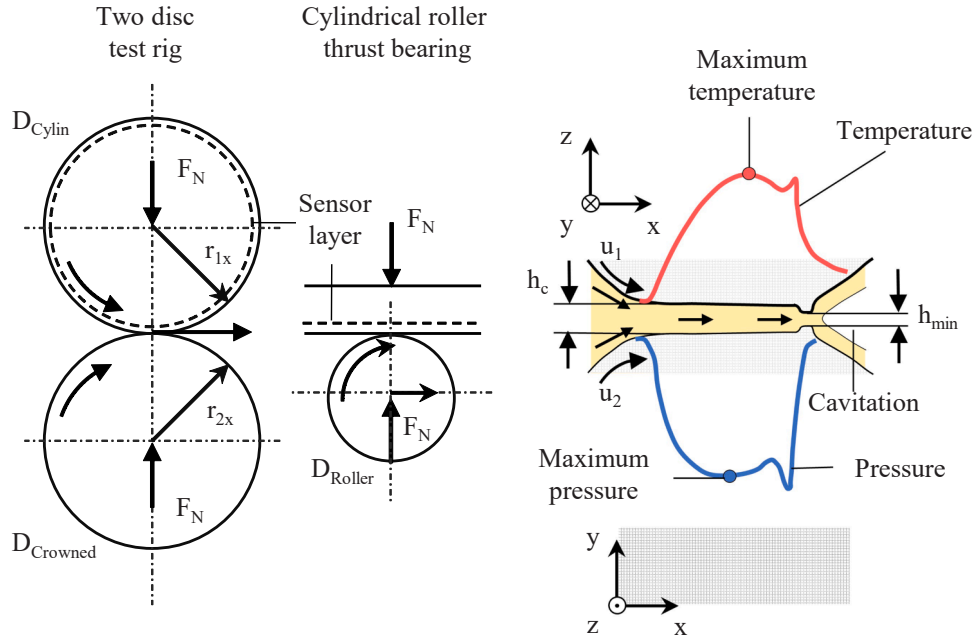


Fig. 8. Contact geometry and typical TEHD pressure, gap height and temperature distribution.

$$\frac{\partial}{\partial x} \left(\Phi_{xx}^p G_{1x} \frac{\partial p_h}{\partial x} \right) + \frac{\partial}{\partial y} \left(\Phi_{yy}^p G_{1y} \frac{\partial p_h}{\partial y} \right) = \frac{\partial}{\partial x} \left(\theta G_{2x} \frac{(u_2 - u_1)}{F_0} + \theta G_3 u_1 + \theta G_3 \Phi_{xx}^s \frac{(u_2 - u_1)}{2h} \right) + \frac{\partial}{\partial t} (\theta G_3) \quad (5)$$

with $p_h > p_{cav}$ and $\theta = 1$ in the pressure zone
 $p_h = p_{cav}$ and $\theta < 1$ in the cavitation zone

$$G_{1x} = G_{1y} = \int_0^h \rho(z) \left[\int_0^z \frac{z'}{\eta^*(z')} dz' - \frac{F_1}{F_0} \int_0^z \frac{1}{\eta^*(z')} dz' \right] dz$$

$$G_{2x} = \int_0^h \rho(z) \left[\int_0^z \frac{1}{\eta^*(z')} dz' \right] dz$$

$$F_0 = \int_0^h \frac{1}{\eta^*(z)} dz \quad F_1 = \int_0^h \frac{z}{\eta^*(z)} dz \quad G_3 = \int_0^h \rho(z) dz$$

The validity of the Reynolds equation applies to the pressure and cavitation zone while ensuring mass conserving flow in the entire lubrication gap. In the pressure zone, where hydrodynamic pressures exceed the cavitation pressure p_{cav} , the gap is entirely filled with fluid. Thus, the gap fill factor θ is set to one. In contrast, in the cavitation zone the gap is no longer completely filled with fluid, resulting in a gap fill factor lower than one. The hydrodynamic pressure is then set to the cavitation pressure.

The equations used to describe the temperature and pressure dependence of the fluid's density and viscosity are presented in Table 7. In this study, the Bode [39] and Rodermund [40,41] models were used, with the parameters for the reference oil FVA 3 also provided. It is typical for lubricants to exhibit non-Newtonian behavior under high pressure and high shear rates, which is characteristic of TEHD contacts. Under these conditions the lubricants do not show a linear Newtonian but a pseudoplastic flow behavior, i.e. the force that can be transmitted via the fluid only increases degressively with increasing shear rate. This

is generally referred to as shear thinning and there are a number of model equations that describe this behavior [42–50]. In this study, the well-known Eyring model was used to describe the shear thinning behavior of the fluid, with temperature and pressure dependent parameters derived from multiple rig tests following the approach outlined by Neupert [51]. The equations used and the corresponding parameters for the reference oil FVA 3 can be found in Table 7.

The film thickness results at each point (x, y) in the local Cartesian coordinate system of the contact (see Fig. 8) from the contours of the contacting bodies z_1 and z_2 , the elastic deformation of the bodies based on the elastic half-space and an integration constant h_0 :

$$h(x, y) = h_0 + z_1(x, y) + z_2(x, y) + \frac{1}{\pi E_{red}} \iint_{(\Omega)} \frac{p_h(x', y') dx' dy'}{\sqrt{(x - x')^2 + (y - y')^2}} \quad (6)$$

To capture the reduced stiffness at the side edges of the contacting bodies, these areas are considered as elastic quarter-space [52]. The equilibrium of forces between the applied and calculated load must met the following condition:

$$F = \int_{(\Omega)} p_h(x, y) d\Omega \quad (7)$$

3.2. Mixed friction

Mixed friction arises when external forces in the contact area transmitted not only through the lubricant but also through solid contacts due to the rough surfaces in the contact. The transition to mixed friction can be defined by a critical gap height h_{cr} , below which solid contacts begin to form between the roughness asperities. The term mixed friction is used when there is still a significant hydrodynamic

Table 7

Description of fluid properties for the reference oil FVA 3.

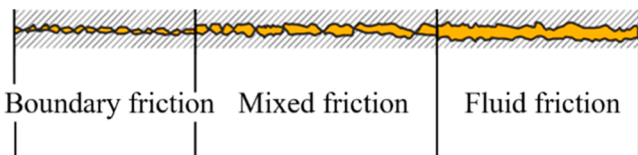
Property Model	Equation	Parameters
Pressure-temperature-dependency of density Bode [39]	$\rho(\vartheta, p_h) = \frac{\rho_0(1 - \alpha_T \vartheta)}{1 - B_0 \ln \left(\frac{B_1 e^{B_2 \vartheta} + p_h}{B_1 e^{B_2 \vartheta} + p_0} \right)}$	$\rho_0 = 1043.23 \text{ kg/m}^3$ $\alpha_T = 0.000527 \text{ 1/K}$ $B_0 = 0.07813$ $B_1 = -3627.93 \text{ bar}$ $B_2 = -0.003419 \text{ 1/K}$ $p_0 = 0 \text{ bar}$
Pressure-temperature-dependency of viscosity Vogel [53]-Rodermund [40,41]	$\eta(\vartheta, p_h) = A e^{\frac{B}{C+\vartheta}} \left(1 + \frac{p_h}{p_0} \right)^{D+E \frac{B}{C+\vartheta}}$	$A = 0.0602 \text{ mPas}$ $B = 986.95 \text{ }^\circ\text{C}$ $C = 98.17 \text{ }^\circ\text{C}$ $D = 0.64$ $E = 0.0097$ $p_0 = 221.98 \text{ MPa}$
Non-Newtonian flow behavior Eyring [51]	$\dot{\gamma}(\tau) = \frac{\tau_c}{\eta} \sinh \left(\frac{\tau}{\tau_c} \right)$ $\tau_c(\vartheta, p_h) = \tau_{c0} + a(\vartheta - \vartheta_0) + b p_h$	$\tau_{c0} = 1 \text{ N/mm}^2$ $a = 0.045 \text{ N/(mm}^2\text{K)}$ $b = 0.0022$ $\vartheta_0 = 0 \text{ }^\circ\text{C}$
Thermal conductivity Power function	$\lambda(\vartheta, p_h) = a_1(1 - a_2 \vartheta) + b_1 \left(\frac{p}{p_0} \right)^{b_2}$	$a_1 = 0.137 \text{ W/(mK)}$ $a_2 = 0.00026 \text{ 1/K}$ $b_1 = 0.00028 \text{ W/(mK)}$ $b_2 = 0.5664$ $p_0 = 1 \text{ bar}$
Specific heat capacity Polynomial	$c_p(\vartheta, p_h) = c_{p,00} + c_{p,t1} \vartheta + c_{p,t2} \vartheta^2 + c_{p,p1} p_h + c_{p,p2} p_h^2 + c_{p,tp} \vartheta p_h$	$c_{p,00} = 1.74 \text{ kJ/(kgK)}$ $c_{p,t1} = 2.4910^{-3} \text{ kJ/(kgK}^\circ\text{C)}$ $c_{p,t2} = -1.17 \cdot 10^{-6} \text{ kJ/(kgK}^\circ\text{C}^2)$ $c_{p,p1} = -3.1 \cdot 10^{-5} \text{ kJ/(kgKbar)}$ $c_{p,p2} = 1.36 \cdot 10^{-12} \text{ kJ/(kgKbar}^2)$ $c_{p,tp} = -8.58 \cdot 10^{-8} \text{ kJ/(kgK}^\circ\text{Cbar)}$

load-bearing effect. In contrast, boundary friction is often referred to when solid contacts dominate (Fig. 9).

The ideal approach to capturing mixed friction is to map the rough surfaces directly in the lubrication gap of the TEHD contact using a solid contact model. However, the resolution of the calculation area required for this is not feasible with the computing technology currently available. As already used by the authors in [21] or more in detail in [33, 54–56], there are analytical equations which define the mixed friction condition as a specific film thickness. But these approaches are rather imprecise, as they do not consider the local roughness. In the simulation model employed in this study, an approach was applied whereby local roughness is considered without direct mapping into the lubrication gap of the TEHD contact [57]. Using a discrete elastic-plastic solid contact model, local solid contact pressure distributions of representative sections of the contact surfaces are first determined for a certain number of contact positions in a preliminary calculation step (Fig. 10). This yields a solid contact pressure curve, representing the progression of the integral solid contact pressure $\bar{p}_c$ over the deformed gap height h_{def} (Fig. 10). The aforementioned curve can be employed to calculate the integral solid contact pressure at each point (x,y) in the lubrication gap of the TEHD contact as a function of the local film thickness. In the case of mixed friction assuming proportional load sharing, the pressure distribution is thus the sum of hydrodynamic and integral solid contact pressures:

$$p(x, y) = p_h(x, y) + \bar{p}_c(x, y) \quad (8)$$

Similarly, the distribution of frictional shear stress can be derived from the hydrodynamic and solid contact shear stresses, assuming a constant boundary friction coefficient f_c :

**Fig. 9.** Definition of fluid, mixed and boundary friction.

$$\tau_f(x, y) = \tau_{fh}(x, y) + \bar{\tau}_{fc}(x, y) \quad (9)$$

with $\bar{\tau}_{fc}(x, y) = f_c \bar{p}_c(x, y)$

The equilibrium of forces then results to:

$$F = \int_{(\Omega)} p(x, y) d\Omega = \int_{(\Omega)} (p_h(x, y) + \bar{p}_c(x, y)) d\Omega \quad (10)$$

Subsequently, the level of mixed friction can be evaluated by introducing a fluid and a solid contact load ratio:

$$L_{FI} = \frac{\int_{(\Omega)} p_h(x, y) d\Omega}{\int_{(\Omega)} p(x, y) d\Omega} 100\% \quad (11)$$

$$L_{SO} = \frac{\int_{(\Omega)} \bar{p}_c(x, y) d\Omega}{\int_{(\Omega)} p(x, y) d\Omega} 100\%$$

If the minimum film thickness h_{min} is below the critical gap height h_{cr} , the solid contact load ratio increases as the film thickness decreases. In turn, the fluid contact load ratio decreases with the same amount.

Film thickness in the range of surface roughness require the consideration of microhydrodynamics. To address this, the Reynolds Eq. (5) is extended by two pressure flow factors Φ_{xx}^p and Φ_{yy}^p and one shear flow factor Φ_{xy}^s . The flow factors are also calculated numerically, considering the local roughness in a preliminary calculation step, similar to the preliminary solid contact simulation. A more detailed description of the model used is given in [57].

Fig. 11 shows exemplary the integral solid contact pressure curve and the pressure flow factors for the polished-grinded contact of the two-disk test rig, determined for the rough surface sections shown in Table 3 and the elastic-plastic properties shown in Table 4. The integral solid contact pressure increases continuously with decreasing gap height and reaches a value of almost 6000 MPa. As a consequence of the grinding process of the counter disk and the resulting anisotropic orientation of

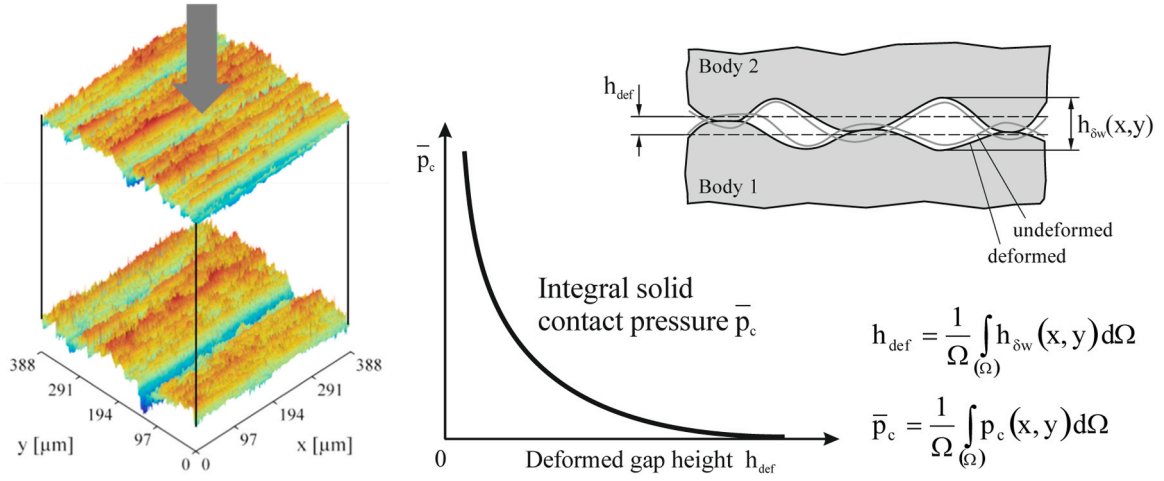


Fig. 10. Solid contact model [35].

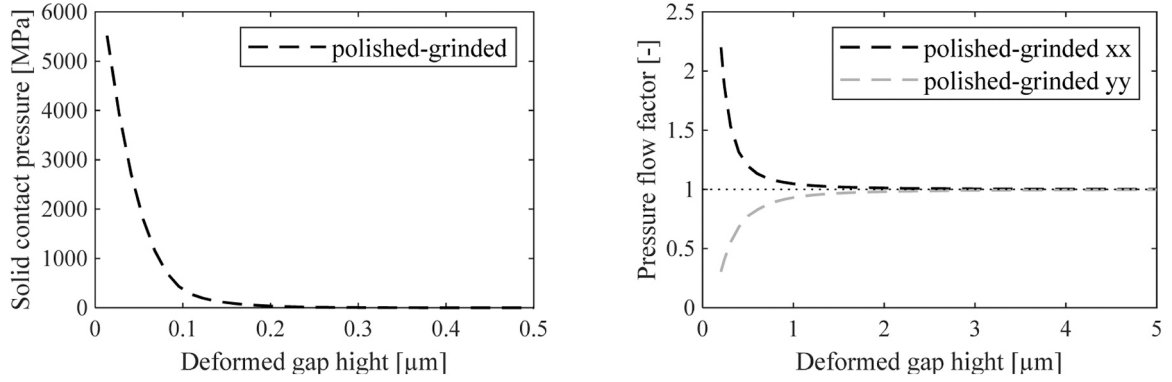


Fig. 11. Two-disk test rig, integral solid contact pressure (left) and pressure flow factor (right).

the roughness, the pressure flow factor decreases in the x direction and increases in the y direction as the gap height decreases. Pressure flow factors less than one will lead to an increase in the hydrodynamic effect, while factors greater than one to a decrease. The boundary friction coefficient was set for all calculations to $f_c = 0.07$.

3.3. Temperature

The temperature calculation is based on the balancing of heat sources and heat sinks. Fluid friction, boundary friction within the solid

contacts and compression heating act as heat sources, while heat conduction, convection and expansion as heat sinks. The heat conduction in the lubrication gap is assumed to be isotropic. The temperature distribution in the lubrication gap is calculated by numerically solving the three-dimensional energy equation:

$$\begin{aligned} & \beta_{\text{th}} \left(\frac{\partial p_h}{\partial t} + u \frac{\partial p_h}{\partial x} + v \frac{\partial p_h}{\partial y} \right) + \tau_{\text{fthx}} \frac{\partial u}{\partial z} + \tau_{\text{fthy}} \frac{\partial v}{\partial z} \\ & = c_p \rho \left(\frac{\partial \theta}{\partial t} + u \frac{\partial \theta}{\partial x} + v \frac{\partial \theta}{\partial y} + w \frac{\partial \theta}{\partial z} \right) - \lambda \left(\frac{\partial^2 \theta}{\partial x^2} + \frac{\partial^2 \theta}{\partial y^2} + \frac{\partial^2 \theta}{\partial z^2} \right) \end{aligned} \quad (12)$$

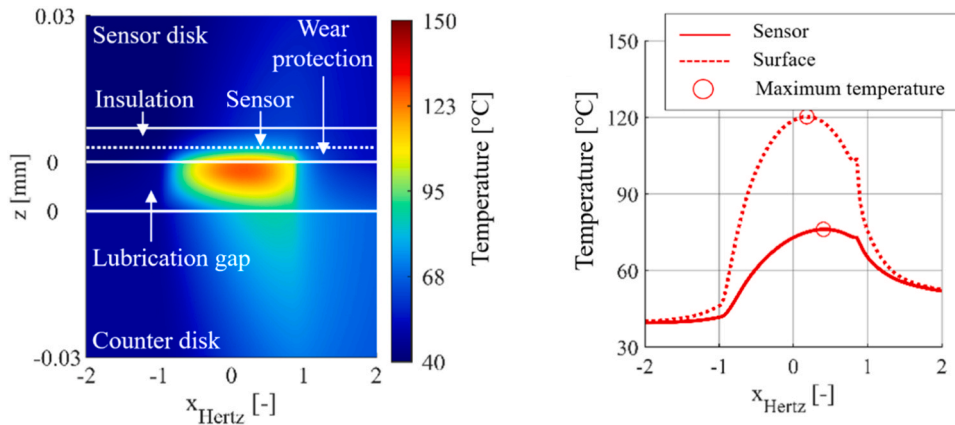


Fig. 12. Exemplary temperature distribution in the undeformed lubrication gap and solid bodies calculated with the TEHD model – left; determined surface and sensor temperature – right; x_{Hertz} – x coordinate normalized with the Hertzian contact width.

The models used to determine the temperature and pressure dependent values for the thermal conductivity λ and the specific heat capacity c_p are given in Table 7. To calculate the temperature distribution in the solids, FOURIER's heat equation is solved:

$$c_{v,solid} \rho_{solid} \left(\frac{\partial \theta}{\partial t} \right) = \lambda_{solid} \left(\frac{\partial^2 \theta}{\partial x^2} + \frac{\partial^2 \theta}{\partial y^2} + \frac{\partial^2 \theta}{\partial z^2} \right) \quad (13)$$

The varying thermophysical material properties of the bodies resulting from the structure of the sensor including the temperature depending thermal conductivity can be precisely taken into account. Further details on the implementation of the temperature calculation can be found in [23,33,34,57].

While the temperature is computed as a 3D field, the sensor temperature measurement relies on the temporal and spatial integration of resistance changes in the conductor, a dynamic quantity. To facilitate a comparison between the measurement and simulation, the calculated temperature field was converted into a scalar quantity in two steps. Initially, the temperature field within the sensor layer was determined. Subsequently, the average temperature along the sensor cross-sections over the sensor length was computed for each discrete point in the x-direction. This represents the mean calculated sensor temperature, enabling a direct comparison with the measured values.

Fig. 12 (left) illustrates an exemplary temperature distribution in a cross-section calculated with the TEHD model. The majority of the heat generated in the contact area due to lubrication shearing is transported through the uncoated disk. However, the maximum lubrication gap temperature is closer to the coated sensor disk due to the thermal insulation effect of the layer. Consequently, the temperature in the wear protection layer and the sensor temperature rises significantly even though there is the aforementioned thermal insulation effect.

Fig. 12 (right) compares the temperature at the surface of the sensor disk with the sensor temperature inside the disk. The maximum temperature at the disk surface reaches a value of 120.3 °C at $x_{Hertz} = 0.18$ and at the sensor a significantly lower value of 76.2 °C at $x_{Hertz} = 0.4$. This shows that in addition to the lower maximum temperature, there is also a depth dependent shift of the maximum temperature in the

direction of the contact outlet, which must be taken into account when comparing experimental and simulation results.

3.4. Numerical procedure

To solve the TEHD problem, the contact is first divided into a defined number of discrete elements in the three spatial directions (x,y,z). For given load, velocity and temperature boundary conditions and under consideration of the preliminary calculated solid contact pressure curve and flow factors the thermal elastohydrodynamic calculation is carried out iteratively by solving the REYNOLDS equation and the energy and FOURIER heat equation one after the other, until the equilibrium of forces and temperatures is reached.

The equations for calculating the elastic deformations of the bodies, the non-Newtonian flow behavior of the lubricant and the mass conserving cavitation are solved simultaneously. The equations for calculating the elastic deformations of the bodies and the solid contact pressures as well as for the temperature and pressure dependent fluid properties and the non-Newtonian flow behavior are solved simultaneously. The main steps of the simulation process are shown in the numerical solution scheme in Fig. 13. A high level of numerical efficiency is achieved by consistent application of multigrid methods [58].

3.5. TEHD pressure distribution and contact temperature

The investigated contact conditions are shown as examples for the two-disk test rig and the cylindrical roller thrust bearing in Fig. 14. The left column shows the pressure distribution and the right column the contact temperature on the surface of the sensor disk resp. washer. The shape differences of the distributions can be seen. The disk geometry of the two-disk contact results in the formation of an elliptical contact area due to the crowning of the counter disk. This indicates that the maximum pressure and temperature, in conjunction with the constant slip, are located at the center of the contact surface. The sensor was consequently positioned in the center of the contact area. The roller washer contact of the cylindrical roller thrust bearing results in a line contact. This pairing causes an effect that leads to significant stress increases at the start and end of the contact line. Therefore, the rollers have a radius at the edge and a logarithmically shaped crowning. The real manufactured contour of the roller in the direction of the length (y direction) was considered in the calculations. Although the rollers have this profile, the maximum pressure is still close to the edge of the contact area. The kinematic slip, which increases towards the edge, and the nearly constant pressure results in an increase in frictional power and this leads in a temperature distribution with two local maxima at the inner and outer raceway radius. Thus, the sensor was positioned at the outer edge of the raceway to measure the highest possible pressure and temperature values.

4. Test results in comparison with TEHD results

4.1. High spatial measurements with the two-disk test rig

The presented sensor enables the simultaneous measurement of both contact temperature and pressure. This approach allows the verification of the exact contact conditions and the correction of a pressure error in temperature measurements. Fig. 15 shows the pressure distribution at three normal forces for a selected slip operating point. Prior to the sensor's entry into the contact at $x_{Hertz} = -1$, the pressure in the inlet area begins to increase. This is followed by a rapid rise to a peak value at $x_{Hertz} \approx 0$, after which a decrease is observed. The common second EHD pressure peak is not identified in the measurements, but there is a slight discontinuity after with the pressure drops sharply and reaches a value of 0 MPa at $x_{Hertz} = 1$. A quantitative comparison of the maximum pressure derived from the TEHD simulation reveals that there are deviations with increasing normal force of 78 MPa, 48 MPa, and 64 MPa,

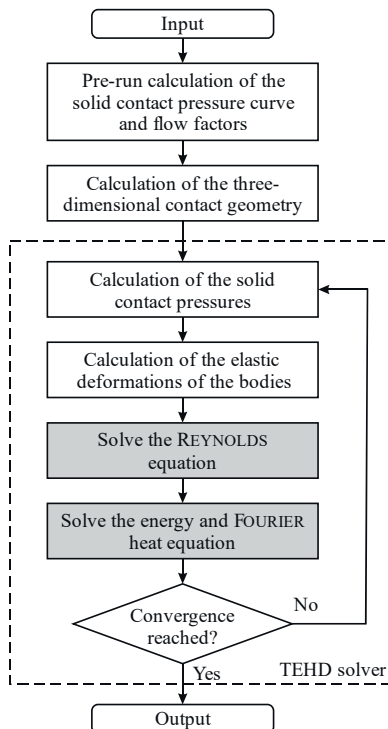


Fig. 13. Numerical solution scheme of the TEHD simulation.

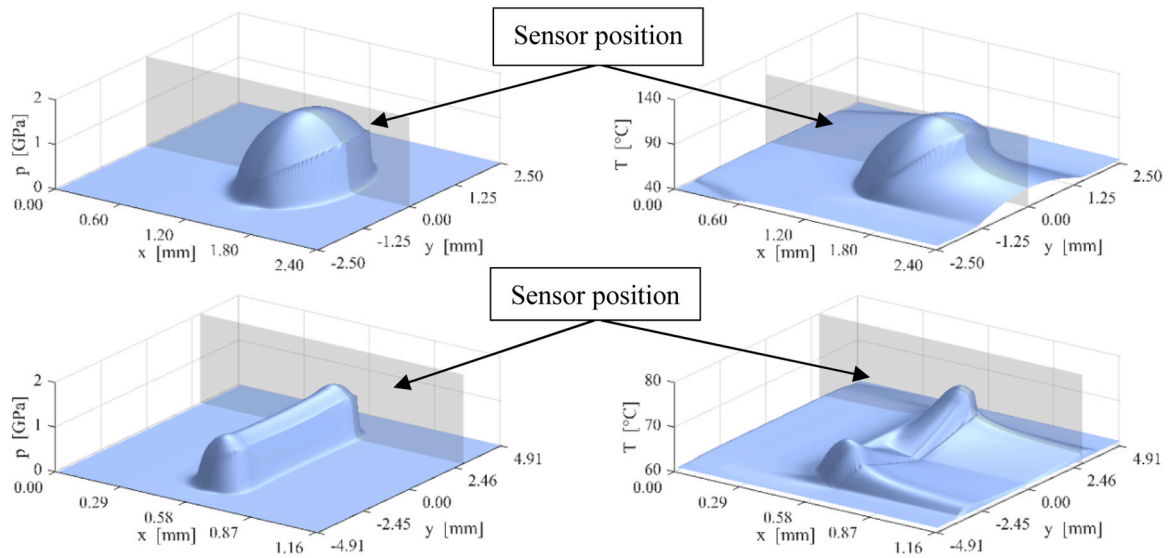


Fig. 14. Examples of pressure distribution and contact temperature of the two-disk test rig – top; and cylindrical roller thrust bearing – bottom.

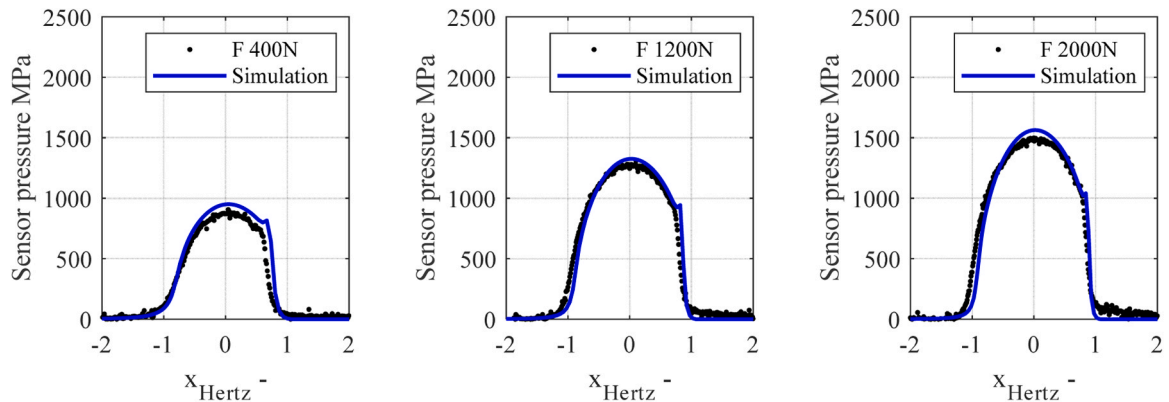


Fig. 15. Pressure measurement at normal force increase with SRR = 22 %, $u_{\text{Rol}} = 2$ m/s, $T_{\text{Oil}} = 40$ °C at $p_{\text{Hmax}} = 1009$ MPa, $p_{\text{Hmax}} = 1368$ MPa and $p_{\text{Hmax}} = 1600$ MPa.

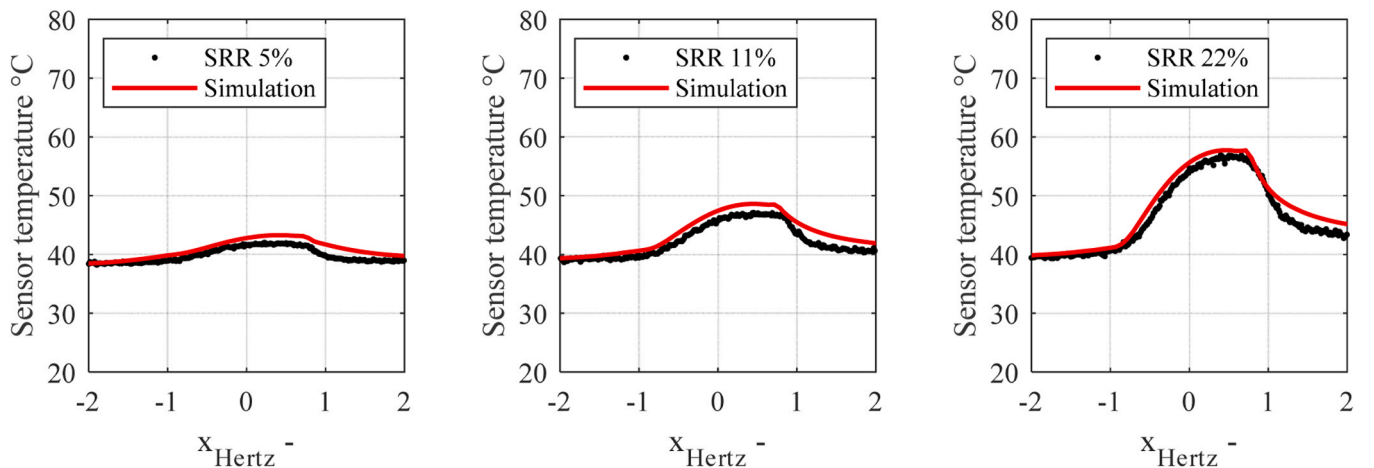


Fig. 16. Temperature measurements for fluid friction with slip increase at $F_N = 400$ N, $p_{\text{Hmax}} = 1009$ MPa, $u_{\text{Rol}} = 2$ m/s, $T_{\text{Oil}} = 40$ °C.

respectively.

Fig. 16 compares the measured values of the sensor temperature for fluid friction for three slip operating points under otherwise identical conditions. Due to the increasing lubrication gap temperature with increasing slip, the central film thickness h_c decreases from $0.843 \mu\text{m}$ to

$0.804 \mu\text{m}$ to $0.754 \mu\text{m}$, but the solid contact ratio still remains at $L_{\text{So}} = 0$ % which implies fluid friction (Table 8). In general, the curves from measurement and simulation show very good agreement. The temperature comparison in the contact middle at $x_{\text{Hertz}} = 0$ yields absolute deviations between the experimental data and the simulation of 1.19 K,

Table 8

Calculated central film thickness and solid contact ratio for fluid friction and mixed friction for three slip operating points.

Central film thickness h_c / solid contact ratio L_{So}	SRR		
	5%	11%	22%
Fluid friction (Fig. 16)	0.843 μm / 0.00%	0.804 μm / 0.00%	0.754 μm / 0.00%
Mixed friction (Fig. 17)	0.231 μm / 0.64%	0.226 μm / 0.80%	0.216 μm / 1.15%

1.63 K and 1.75 K with increasing slip which only results in small relative deviations from 2,87 %, 3,56 % and 3,24 %.

Fig. 17 compares the temperature measurement with the simulation at higher pressures and with mixed friction. The oil feed temperature was increased and the rolling velocity u_{Rol} of the disks was reduced. This results in calculated lubrication gap heights h_c of 0.231 μm ($L_{So} = 0.64$ %), 0.226 μm ($L_{So} = 0.80$ %) and 0.216 μm ($L_{So} = 1.15$ %) and thus

to mixed friction $L_{So} > 0$ % (Table 8). The sensor temperature in the inlet area of 53 °C is cooler than the oil feed temperature (see Section 5.1). A comparison of temperatures at $x_{\text{Hertz}} = 0$ reveals absolute deviations of 0.96 K, 0.10 K and -2.02 K, and relative deviations of 1.63 %, 0.16 % and -2.74 % as a consequence of increasing slip. At SRR = 22 %, the simulation is in good agreement with the experimental data. However, at SRR = 5 % and SRR = 11 %, the cooling effect of the sensor in the outlet area is slightly higher than predicted by the simulation. In the inlet area, the temperature rise of the sensor has a small delay for both operating points.

The qualitative and quantitative prediction of the temperatures for fluid friction and mixed friction in the contact area, as well as the cooling in the outlet area, deviate only a few Kelvin from the measurements. This indicates that the prediction of the friction in the contact area and the determination of the heat conduction in the disks and in the sensor layers are accurate.

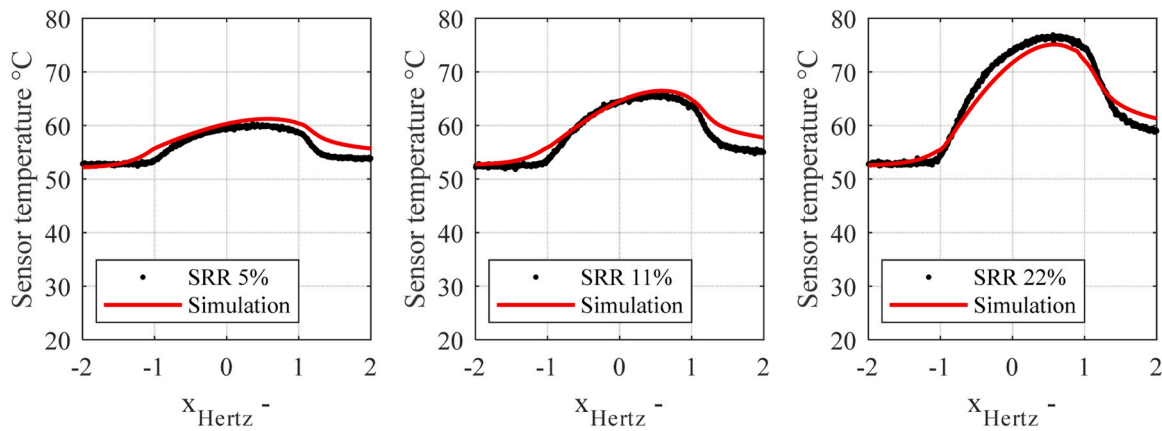


Fig. 17. Temperature measurements for mixed friction with slip increase at $F_N = 1600$ N, $p_{\text{Hmax}} = 1493$ MPa, $u_{\text{Rol}} = 1$ m/s, $T_{\text{Oil}} = 60$ °C.

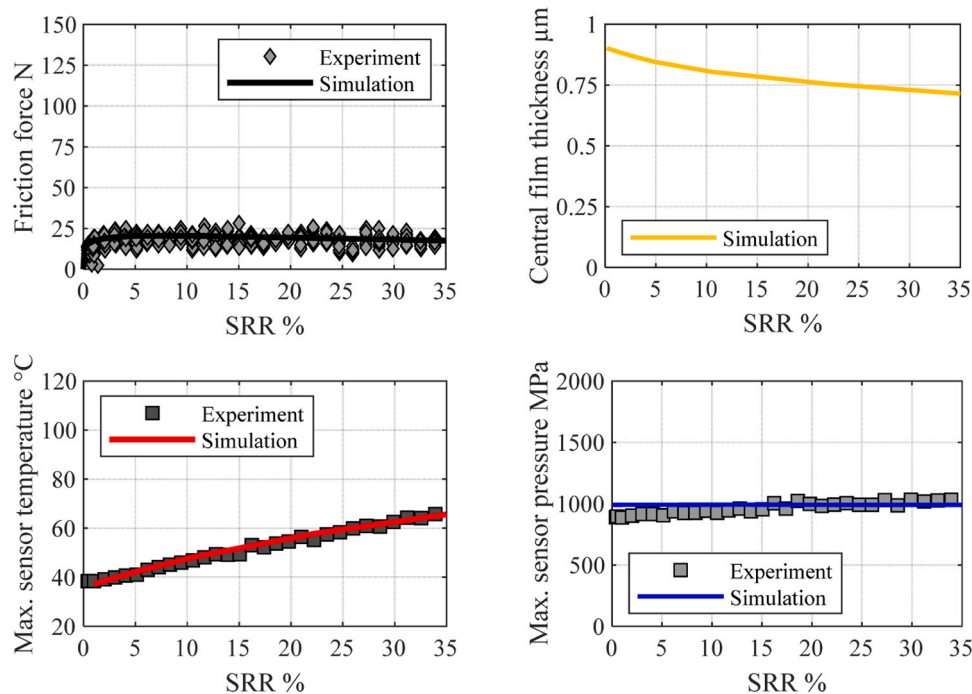


Fig. 18. Traction measurement at fluid friction in comparison to the maximum sensor temperature and pressure and to the calculated central film thickness at $F_N = 400$ N, $p_{\text{Hmax}} = 1009$ MPa, $u_{\text{Rol}} = 2$ m/s, $T_{\text{Oil}} = 40$ °C.

4.2. Traction measurements with the two-disk test rig

In addition to the presented measurements with high spatial resolution, a series of traction measurements were conducted with the objective of studying the fluid behavior in the presence of varying slip conditions. Fig. 18–Fig. 20 present a comparison of the measured and calculated friction forces, maximum sensor temperatures and maximum sensor pressures. The calculated central film thickness is also included for purposes of clarification. For moderate pressures ($p_{Hmax} = 1009$ MPa, Fig. 18), the friction forces can be accurately predicted over the investigated SRR range. The friction forces initially exhibit a pronounced increase within a range of $SRR < 1\%$, after which they remain relatively constant up to $SRR = 35\%$. This fits very well the behavior described in the literature [59]. The maximum sensor temperature can also be predicted with reasonable accuracy in absolute terms and in terms of increase. With the friction remaining almost constant, the friction power increases in proportion to the SRR, which leads to an almost linear increase in the maximum sensor temperature. The maximum sensor pressure appears to increase slightly with increasing pressure, which may be due to a remaining temperature influence on the pressure measurement, even though the temperature influence was also corrected for the pressure measurement. The calculated central film thickness for $SRR \approx 0\%$ is $0.903\ \mu\text{m}$, with a gradual decrease to $0.714\ \mu\text{m}$ at $SRR = 35\%$. This reduction in lubricating film thickness can be explained by the elevated contact temperature and the subsequent decline in viscosity.

When the pressure is increased to a more technically relevant range for rolling contacts ($p_{Hmax} = 1600$ MPa, Fig. 19) the friction again increases strongly up to $SRR < 1\%$ and rises to approximately $SRR \approx 2.5\%$, after which the friction force decreases again. The reduction in friction with increasing slip can be explained by the strong friction-related increase in temperature and the associated reduction in effective viscosity. The maximum friction force is several times higher than at a pressure of 1009 MPa and the measured force is higher than the calculated friction force. The decrease in friction leads to the fact that the maximum sensor temperature no longer rises linearly with increasing slip, but increases with a slower rate and thus flattens out. In contrast to the friction prediction, the temperature progression is very accurately

represented by the calculations. The evaluation of the maximum pressure also shows a slight increase in pressure with increasing slip. The calculated central film thickness for $SRR \approx 0\%$ is $0.777\ \mu\text{m}$, exhibiting a gradual decrease to $0.510\ \mu\text{m}$ at $SRR = 35\%$. This lower film thickness in comparison to that observed in Fig. 18 can be attributed to the typical EHD behavior that with increasing pressure the film thickness decreases.

In a further traction measurement (Fig. 20), the rolling velocity was reduced and the oil feed temperature was increased at $p_{Hmax} = 1600$ MPa. This was done in order to reduce the lubricating film thickness and thus induce mixed friction up to $L_{S0} = 5.15\%$ at $SRR = 35\%$. The friction force increases significantly once more up to $SRR < 1\%$, and reaches its maximum at $SRR \approx 5\%$. The friction force then decreases slightly. However, the simulation again predicts lower friction forces. The analysis of the maximum sensor temperature shows discontinuities, which may be influenced by the speed control. A constant speed is necessary as both temperature and pressure signals must be synchronized via the speed for error compensation. However, the temperature curve shows good agreement with the calculations as the SRR increases. The problems of speed control are also reflected in the pressure measurements.

The comparison of the measured and the calculated temperature data reveals a high agreement for both low and high pressures, as well as across a broad range of slip values. The data demonstrate that the modelling of the fluid behavior is accurate, although there is a discrepancy between the friction measurements and the calculated data at high pressures. The deviations between the friction force measurement and the simulation increase with higher normal force. It is assumed that the influence of the normal force on the friction force measurement is caused by deformations of the test rig at very high loads. The calculated central film thickness for mixed friction is considerably lower than for fluid friction due to the lower rolling velocity (Fig. 18 and Fig. 19). The film thickness is $0.286\ \mu\text{m}$ for $SRR \approx 0\%$ and $0.157\ \mu\text{m}$ for $SRR = 35\%$. The decrease in film thickness is less significant because the increase in contact temperature is lower due to the lower sliding speed.

4.3. Measurements with a cylindrical roller thrust bearing

In a subsequent series of experiments, the pressure and temperature

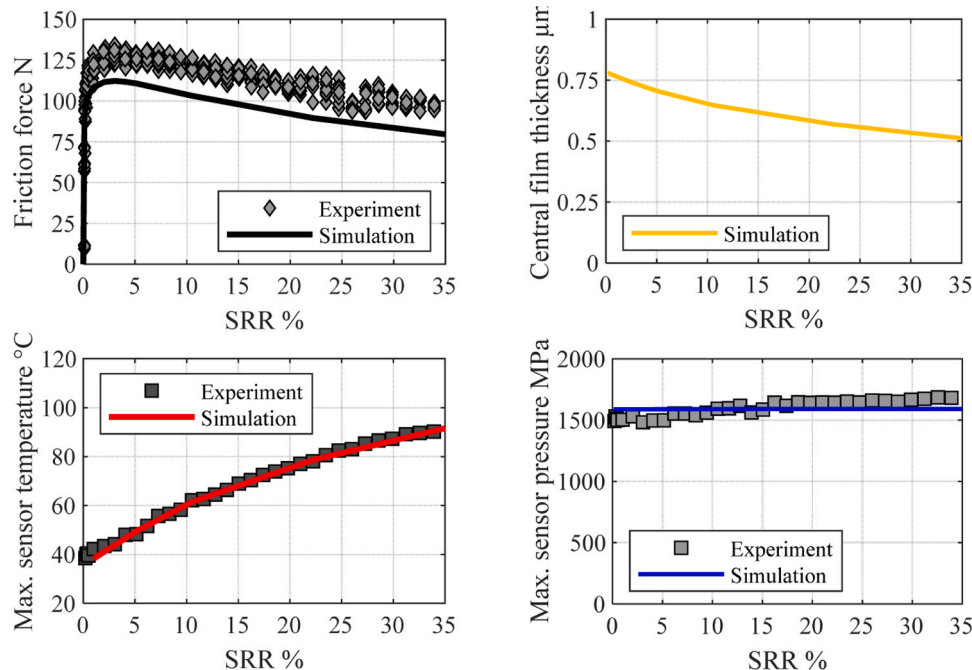


Fig. 19. Traction measurement at fluid friction in comparison to the maximum sensor temperature and pressure and to the calculated central film thickness at $F_N = 2000$ N, $p_{Hmax} = 1600$ MPa, $u_{Rol} = 2$ m/s, $T_{Oil} = 40$ $^{\circ}\text{C}$.

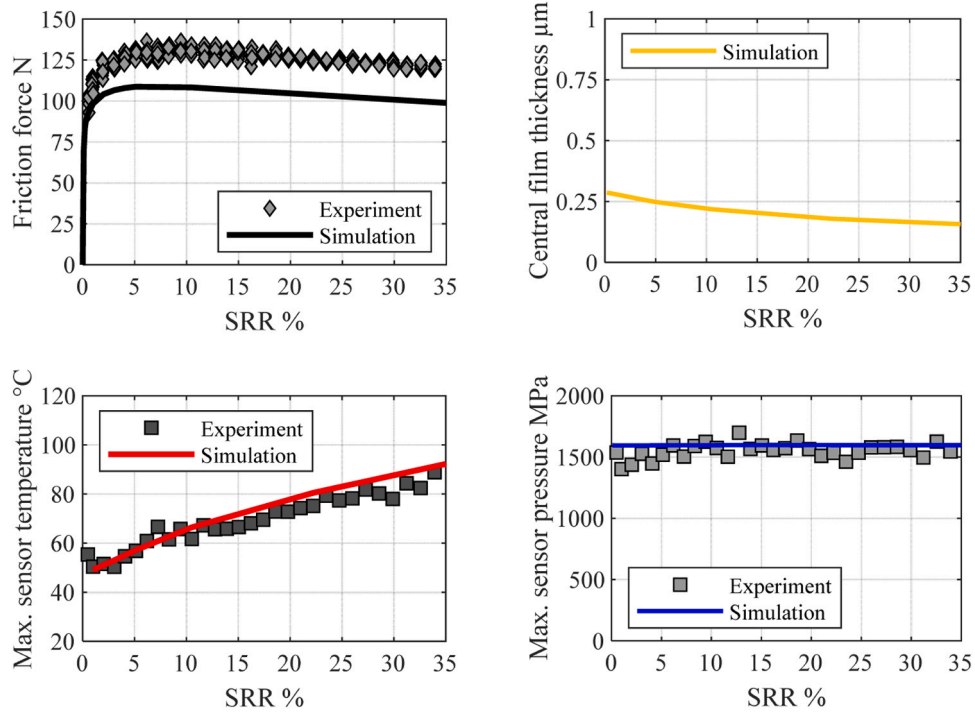


Fig. 20. Traction measurement at mixed friction in comparison to the maximum sensor temperature and pressure and to the calculated central film thickness at $F_N = 2000 \text{ N}$, $p_{H\max} = 1600 \text{ MPa}$, $u_{\text{Rol}} = 1 \text{ m/s}$, $T_{\text{Oil}} = 60 \text{ }^{\circ}\text{C}$.

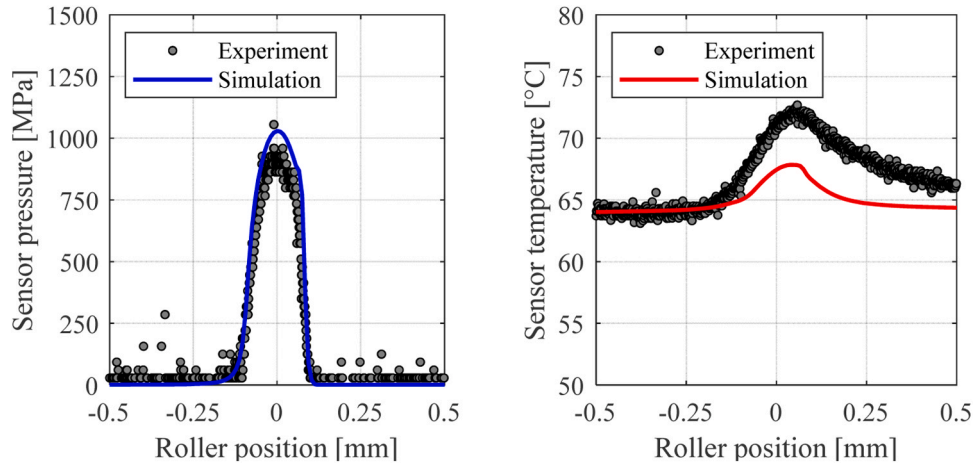


Fig. 21. Pressure and temperature measurement on the 81212 cylindrical roller thrust bearing for a single roller-sensor contact at $F_N = 20 \text{ kN}$, $p_{H\max} = 884 \text{ MPa}$, $n = 1000 \text{ rpm}$, $T_{\text{Bea}} = 60 \text{ }^{\circ}\text{C}$.

sensors were applied to the washer of a cylindrical roller thrust bearing. The housing washer, which contains the sensors, remains stationary while the rolling element is brought into contact with the sensor. The rolling elements transfer the bearing load from the housing washer to the rotating shaft washer. Fig. 21 – left illustrates the measured and calculated pressure distribution with fluid friction. At a roller position of -0.15 mm , the pressure begins to increase due to compression of the fluid in the inlet area. From a distance of -0.10 mm to the center, the pressure then rises significantly. At the roller position of 0.00 mm the maximum pressure of 895 MPa is reached, but the maximum pressure is calculated to be up to 1029 MPa . This indicates that there is a deviation $> 100 \text{ MPa}$ between the measurement and the calculation for the presented roller-sensor contact. Following the maximum pressure, both the simulated and measured pressures decrease significantly to 0 MPa at 0.10 mm .

Fig. 21– right shows the sensor temperature, both as the simultaneous measured value and as a calculated value. Prior to entering the contact area, the sensor has a constant temperature of $65 \text{ }^{\circ}\text{C}$, despite the washer temperature being maintained at $60 \text{ }^{\circ}\text{C}$. The temperature difference can be attributed to the thermal insulation effect of the insulation layer. The temperature rise already begins in the inlet area due to inlet shear heating and thus before the rolling element position of -0.1 mm which indicates the beginning of slow pressure increase. The temperature then increases to $71.7 \text{ }^{\circ}\text{C}$. The calculated maximum temperature is $67.8 \text{ }^{\circ}\text{C}$, which is below the measured value. It is notable that also the cooling curve exhibits a considerable deviation. The calculations indicate that the temperature will rapidly decline to its initial value of $65 \text{ }^{\circ}\text{C}$. However, the measured cooling curve is significantly less steep. The possible causes are discussed in Chapter 5 and require further investigation, particularly through additional measurements.

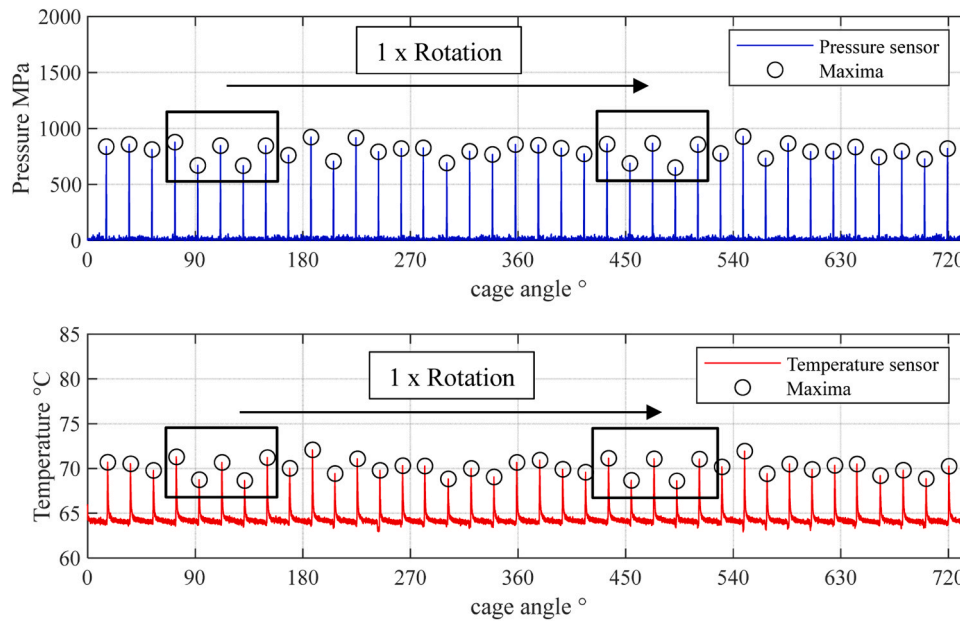


Fig. 22. Pressure and temperature measurement on the 81212 cylindrical roller thrust bearing for two cage rotations at $F_N = 20$ kN, $p_{Hmax} = 884$ MPa, $n = 1000$ rpm, $T_{Bea} = 60$ °C.

The cage is composed of 19 rollers. The measurement presented in Fig. 21 was selected for examination as a representative example of a single roller-sensor contact. Fig. 22 illustrates the measurement obtained from two cage rotations. Each peak represents a single roller-sensor contact, and the maxima in pressure and temperature are additionally marked for enhanced comparison. It can be observed that the signal pattern is repeated for each rotation. Of particular note is the similarity between the signal patterns for the temperature and pressure measurements, which leads to the conclusion that the cylindrical roller thrust bearing has a systemic influence. The maximum pressure varies between the lowest and highest values from 667 MPa to 922 MPa, while the maximum temperature varies between 68.7 °C and 72.0 °C.

5. Discussion

5.1. Two-disk measurements with high spatial resolution

Initially, spatially resolved pressure and temperature measurements were performed. To compensate the pressure and temperature error, the distance between the sensors was initially considered, and then a linear system of equations according to Köhler [6] was solved. This resulted in an improvement in the quality of the measured values compared to the previous study [21].

As illustrated in Fig. 15, the experimental results indicate that a second pressure peak could not be identified with certainty, despite the simulation data showing a slight peak. However, the pressure buildup in the inlet area ($x_{Hertz} < -1$) and the sharp pressure drop following the second pressure peak were detected. Similarly, the second temperature peak could not be identified with certainty. This is attributed to two main causes: the disk design and the sensor volume. The disk design results in a relatively modest second pressure and temperature peak, as illustrated in the simulation in Fig. 12 (top). This is further reduced by the measurement of the mean sensor temperature across the sensor length, width and height.

Mixed friction cannot be detected directly with the sensor but could only be determined under harsh conditions via surface changes, e.g. abrasion or delamination of the layers or by a prediction, that can be made mathematically via the specific lubricant film thickness [33, 54–56] or in the TEHD via the solid contact ratio [23,33,57].

A comparison of the temperatures in fluid friction (Fig. 16)

demonstrates a high degree of accordance between the calculated and experimental results, indicating that the dissipation of shear stress to frictional heat and the thermal conduction effect were accurately determined. The cooling effect in the outlet area is slightly higher in the experimental data than in the simulation. Cavitation could affect the temperature in the outlet region. The cavitation model used in the simulation is already mass conserving, but still simplified by assuming homogeneous two-phase flow in the cavitation zone. This could have a slight impact on the calculated heat transfer and friction and thus on the temperature distribution in the cavitation zone. In addition, temperature boundary conditions affect cooling, but these were assumed to be constant along the entire contact.

Similarly, in the context of mixed friction, there is also a notable degree of accordance (Fig. 17), but with deviation in the inlet and outlet area. The deviation of the cooling effect in the outlet area could be due to cavitation or the constant temperature boundary conditions, as described above. The divergence between the experimental and calculated temperature data in the inlet area may be attributed to shear heating at the inlet area. This phenomenon is more pronounced under mixed friction conditions at very low film thicknesses. It is conceivable that the temperature boundary conditions exert a greater influence under very low film thickness. Moreover, there is a deviation between the temperature of the sensor prior to the contact area and the oil feed temperature. As the oil temperature rises, the temperature difference to the remainder of the test setup also increases. It is a possibility that the cooling effect of the disc and shaft is more pronounced. An additional temperature measurement in the steel part of the sensor disk (below the sensor layers) would provide further insight, but the number of sensor signals that can be transmitted was limited.

5.2. Two-disk traction measurements

To gain further insight into the operating points and confirm the temperature comparison, traction measurements were conducted and the friction forces were recorded. The comparison of the maximum temperatures from these measurements revealed a high degree of correlation with the simulated values, suggesting that the calculations for the lubrication gap and solids are accurate.

At elevated pressures or forces, the friction force measurements indicate a qualitative alignment with the calculated values, but the

observed values tend to exceed the simulation after the initial rise at $SRR \approx 0$. Due to the high forces in conjunction with the large lever arms, the test rig deforms, which could result in a height offset of the disks and, consequently, a normal force influence on the friction force measurement. The pressure measurements indicate a slight increase with increasing slip, despite the pressure or force remaining constant. The pressure measurement was corrected for the temperature error; however, a potential dependence of the pressure coefficient α_p on the temperature was not considered, which could explain the observed slight increase.

Fig. 20 illustrates a traction curve with mixed friction, exhibiting a notable degree of irregularity. The pressure and temperature measurements are conducted at a distance of 0.8 mm, requiring synchronization of both signals. This is achieved through the use of speed, which is assumed to be a constant test parameter in the evaluation. It is important to note, however, that larger speed deviations may occur at very low speeds with the used engines, which could potentially lead to errors in the compensation process.

Occasionally, overloading can lead to total failure of the sensors or a change in material properties. In situations where the load was exceedingly high, it was sometimes necessary to recalibrate i.e. the chromium sensor. Overloading can result in alterations to the plastic material changes within the sensor, which in turn affect the electrical resistance [60]. Also, mixed friction can lead to abrasion or delamination of the wear protection layer, which can lead to an exposure of the sensor. The sensor will continue to function, but a change in the output resistance must then be expected when using manganin. One possible cause may be oxidation under atmospheric oxygen [7,16].

5.3. FE8 rolling bearing tests

Tests were carried out with the FE8 rolling bearing test rig in accordance with the DIN 51819-1 standard using the presented sensor technology.

The width of the pressure distribution, as well as the curve profile and maximum value, align closely with the calculated results. This indicates that the kinematic conditions of the rolling elements were accurately captured in the calculation. The deviations in maximum pressure values concerning different rollers may have been caused by the profiling or by the pocket position in the cage.

In comparison to the pressure, the temperature measurements deviate more significantly from the simulation. The maximum temperatures determined and the cooling curve do not align with the simulation results in terms of both qualitative and quantitative characteristics. In accordance with the specifications outlined in DIN 51819-1, the FE8 test rig is equipped with two temperature measuring points on the housing washer side, which are also used for keeping the temperature constant for the test run. The viscosity is in turn dependent on the temperature in the contact area, which is influenced by the oil feed temperature. This has a significant impact on the ability to accurately predict the friction-related temperature increase. However, an additional temperature measurement on the test head affects the measurement of the friction torque. The rolling element temperature represents another challenging parameter to predict, given that it cannot currently be measured directly in the test with a reasonable technical effort. For instance, the heating effect resulting from the friction between the roller and cage has not been considered yet. Nevertheless, the temperature of the rolling element influences the temperature of the lubricant in the contact area.

6. Conclusions

Thin-film sensors were successfully applied to disks and bearing washers and the pressure as well as the temperature distribution could

be measured under fluid and mixed friction conditions. The following conclusions can be drawn from this study:

1. In comparison to the previous study [21], the coating system underwent further mechanical optimization with regard to the stresses experienced in a rolling contact. This enabled the measurement of the sensor performance under varying loads of up to 1600 MPa conditions for over 10^6 over-rollings, representing a significant increase of one order of magnitude. Furthermore, in preliminary tests, the wear protection layer employed demonstrated a 20 % higher load capacity than the previous wear protection layer.
2. The measurement data demonstrate a high degree of agreement between the two-disk tests and the simulation. A comparison of temperature measurements with simulations can only be successful if the thermal properties of the sensor materials, lubricant and boundary conditions are known precisely and the pressure error is corrected. For this reason, the thermal conductivities and layer thicknesses of the sensor materials were determined. As shown, thermal conductivity depends on temperature. Therefore, the material parameters should be modeled as a function of temperature. Additionally, the exact thickness of the wear protection layer must be known, as there is a strong temperature gradient in the layers.
3. As a lubricant, a very well-studied reference oil was selected and measured. Precise characterization is essential if temperatures and friction forces are to be compared. The inlet temperature of the lubricant and the solid temperature must be known, as they affect not only the temperature measurement but also the friction.
4. The pressure measurements are more directly comparable with the calculations, as the forces are transmitted immediately to the sensor. However, the pressure measurement is crucial for compensating for the error compensation of the temperature measurement.
5. The transfer to component tests is a more challenging process, requiring the implementation of advanced test equipment in accordance with the standard specifications.

Future investigations with the FE8 rolling bearing test rig will include measurements at different loads, for which sensors have already been set up at the calculated positions of maximum temperature increase. The tests will be performed under mixed friction conditions and the longevity of the sensors will be evaluated. Furthermore, the sensor is to be extended to include the measurement of lubrication gap height. It is intended that an extension of the sensor will be developed to include the measurement of lubrication gap height, thus enabling the determination of the friction conditions at a later stage.

CRediT authorship contribution statement

Conceptualization, S. Emmrich and M. Plogmeyer; Data curation, S. Emmrich and M. Plogmeyer; Formal analysis, S. Emmrich; Investigation, S. Emmrich; Software, L. Bobach; Project administration, S. Emmrich and M. Plogmeyer; Resources, D. Bartel and C. Herrmann; Supervision, D. Bartel and A. Schott; Validation, S. Emmrich and L. Bobach; Visualization, S. Emmrich; Writing – original draft, S. Emmrich and M. Plogmeyer; Writing – review & editing, L. Bobach, D. Bartel, A. Schott and C. Herrmann; All authors have read and agreed to the published version of the manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data Availability

Data will be made available on request.

References

- [1] Orcutt FK. Experimental study of elastohydrodynamic lubrication. *A S L E Trans* 1965;8(4):381–96. <https://doi.org/10.1080/05698196508972109>.
- [2] Schouten M. Einfluss elastohydrodynamischer Schmierung auf Reibung, Verschleiss und Lebensdauer von Getrieben [Dissertation]. Eindhoven. Eindhoven: Technische Hogeschool; 1973.
- [3] Kannel JW, Zugaro FF, Dow TA. A method for measuring surface temperature between rolling/sliding steel cylinders. *J Lubr Technol* 1978;100(1):110–4. <https://doi.org/10.1115/1.3453098>.
- [4] Safa M, Leather JA, Anderson JC. Thin film microtransducers for elastohydrodynamic lubrication studies. *Thin Solid Films* 1979;64(2):257–62. [https://doi.org/10.1016/0040-6090\(79\)90518-2](https://doi.org/10.1016/0040-6090(79)90518-2).
- [5] Safa M, Anderson JC, Leather JA. Transducers for pressure, temperature and oil film thickness measurement in bearings. *Sens Actuators* 1982;3:119–28. [https://doi.org/10.1016/0250-6874\(82\)80013-9](https://doi.org/10.1016/0250-6874(82)80013-9).
- [6] Köhler A. Die Entwicklung von aufgedampften Messwertaufnehmern und deren Anwendung zur Druck- und Temperaturmessung in geschmierten Wälz- und Gleitkontakten (engl. The development of vapor-deposited transducers and their application for pressure and temperature measurement in lubricated rolling and sliding contacts) [Dissertation]. Aachen: T. H. Aachen; 1981.
- [7] Baumann H. Druck- und Temperaturmessungen mittels aufgedampfter Dünnschichtaufnehmer in einem elastohydrodynamischen Linienkontakt [Dissertation]. Karlsruhe: Universität Karlsruhe; 1985.
- [8] Baumann H. Measuring surface temperatures between rolling steel cylinders using double-layer transducers. *Proc Inst Mech Eng, Part C J Mech Eng Sci* 1987;201(4):263–70. https://doi.org/10.1243/PIME_PROC_1987_201_119_02.
- [9] Peeken H, Ayanoglu P. Temperature and pressure measurements in gear contacts with thin-film-transducers;14:161–165. [https://doi.org/10.1016/S0167-8922\(08\)70190-1](https://doi.org/10.1016/S0167-8922(08)70190-1).
- [10] Kagerer E, Königer ME. Ion beam sputter deposition of thin film sensors for applications in highly loaded contacts. *Thin Solid Films* 1989;182(1–2):333–44. [https://doi.org/10.1016/0040-6090\(89\)90269-1](https://doi.org/10.1016/0040-6090(89)90269-1).
- [11] Kagerer E. Messung von elastohydrodynamischen Parametern im hochbelasteten Scheiben- und Zahnkontakt [Dissertation]. München: Technische Universität München; 1991.
- [12] Knauf C. Keramische Werkstoffe unter elastohydrodynamischen Bedingungen [Dissertation]. Karlsruhe: TU Karlsruhe; 1994.
- [13] Holland K. Dünnschicht-Sensoren zur gemeinsamen resistiven Druck- und Temperaturmessung in hochbelasteten geschmierten Kontakten; 1998.
- [14] Kühl S. Gemeinsame kontaktauflösende Messung von Druck und Temperatur im elastohydrodynamischen Linienkontakt kraftübertragender Evolventenzahnräder [Dissertation]. Clausthal: TU Clausthal; 1996.
- [15] Sperrfechter T, Haller R. Elastohydrodynamic lubrication between rolling and sliding ceramic cylinders: an experimental investigation. *J Tribol* 2000;122(4):721–4. <https://doi.org/10.1115/1.1310332>.
- [16] Sperrfechter T. Keramische Bauteile im elastohydrodynamischen Kontakt [Dissertation]. Karlsruhe: Karlsruhe: Universität; 1998.
- [17] Dauber O. Elastohydrodynamische Rollreibung in Stahl-Keramik-Kontakten [Dissertation]. Karlsruhe: Universität Karlsruhe; 2001.
- [18] Linzmayer M. Präsentation Projekt F.V.A. 707 V Risse auf Lagerringen - Ringversuch.
- [19] Mayer J. Einfluss der Oberfläche und des Schmierstoffs auf das Reibungsverhalten im EHD-Kontakt [Dissertation]. München: Technische Universität München.
- [20] Hofmann S, Maier E, Lohner T. In situ contact analysis of polyetheretherketone under elastohydrodynamic lubrication. *Polymers* 2022;14(20). <https://doi.org/10.3390/polym14204398>.
- [21] Emmrich S, Plogmeyer M, Bartel D, Herrmann C. Development of a Thin-Film Sensor for In Situ Measurement of the Temperature Rise in Rolling Contacts with Fluid Film and Mixed Lubrication. *Sens (Basel)* 2021;21(20). <https://doi.org/10.3390/s21206787>.
- [22] Kirchner E, Wallmersperger T, Gwosch T, Menning JDM, Peters J, Breimann R, et al. A review on sensor-integrating machine elements. *Adv Sens Res* 2024;3(4). <https://doi.org/10.1002/adsr.202300113>.
- [23] Beilicke R, Bobach L, Bartel D. Transient thermal elastohydrodynamic simulation of a DLC coated helical gear pair considering limiting shear stress behavior of the lubricant. *Tribol Int* 2016;97:136–50. <https://doi.org/10.1016/j.triboint.2015.12.046>.
- [24] Habbchi W, Bair S. The role of the thermal conductivity of steel in quantitative elastohydrodynamic friction. *Tribology Int* 2020;142:105970. <https://doi.org/10.1016/j.triboint.2019.105970>.
- [25] Reddyhoff T, Schmidt A, Spikes H. Thermal conductivity and flash temperature. *Tribol Lett* 2019;67(1). <https://doi.org/10.1007/s11249-018-1133-8>.
- [26] Laukotka, E., M., Arnold, D. Referenzöle Datensammlung: FVA Heft 660.
- [27] Franke JWH, Fritz J, Koenig T, Merk D. Influence of tribolayer on rolling bearing fatigue performed on an FE8 test rig—a follow-up. *Lubricants* 2023;11(3):123. <https://doi.org/10.3390/lubricants11030123>.
- [28] Reynolds O. On the theory of lubrication and its application to Mr. Beauchamp tower's experiments, including an experimental determination of the viscosity of olive oil. *Philos Trans Roy Soc* 1885;1:157.
- [29] Dowson D, Higginson GR. A numerical solution to the elasto-hydrodynamic problem. *J Mech Eng Sci* 1959;1(1):6–15. https://doi.org/10.1243/JMES_JOUR_1959_001_004_02.
- [30] Hamrock BJ, Dowson D. Isothermal elastohydrodynamic lubrication of point contacts: Part I—theoretical formulation. *J Lubr Technol* 1976;98(2):223–8. <https://doi.org/10.1115/1.3452801>.
- [31] Hamrock BJ, Dowson D. Isothermal elastohydrodynamic lubrication of point contacts: Part II—ellipticity parameter results. *J Lubr Technol* 1976;98(3):375–81. <https://doi.org/10.1115/1.3452861>.
- [32] Crook AW. Elastohydrodynamic lubrication of rollers. *Nature* 1961;190(4782):1182–3. <https://doi.org/10.1038/1901182a0>.
- [33] Bartel D. Simulation von tribosystemen: Grundlagen und anwendungen. 1st ed. Wiesbaden [Germany]: Vieweg+Teubner; 2010.
- [34] Bobach L, Bartel D, Beilicke R, Mayer J, Michaelis K, Stahl K, et al. Reduction in EHL friction by a DLC coating. *Tribol Lett* 2015;60(1). <https://doi.org/10.1007/s11249-015-0594-2>.
- [35] Bobach L, Beilicke R, Bartel D. Transient thermal elastohydrodynamic simulation of a spiral bevel gear pair with an octoidal tooth profile under mixed friction conditions. *Tribology Int* 2020;143:106020. <https://doi.org/10.1016/j.triboint.2019.106020>.
- [36] Tribo-X Tribo Technologies GmbH; (<https://www.tribo-technologies.com/en>); 2024.
- [37] Peiran Y, Shizhu W. A generalized reynolds equation for non-newtonian thermal elastohydrodynamic lubrication. *J Tribology* 1990;112(4):631–6. <https://doi.org/10.1115/1.2920308>.
- [38] Elrod HG. A cavitation algorithm. *J Lubr Technol* 1981;103(3):350–4. <https://doi.org/10.1115/1.3251669>.
- [39] Bode B. Modell zur Beschreibung des Fließverhaltens von Flüssigkeiten unter hohem Druck. *Tribol + Schmier* 1989;36:182–9.
- [40] Rodermund H. Berechnung der Temperaturabhängigkeit der Viskosität von Mineralölen aus dem Viskositätsgrad. *Schmieringstechnik und Tribologie*: p. 56–57; Available from: (<https://pascal-francis.inist.fr/vibad/index.php?action=getrecorddetail&idt=pascal7880352955>).
- [41] Mihailidis A, Retzepis J, Salpistis C, Panajiotidis K. Calculation of friction coefficient and temperature field of line contacts lubricated with a non-Newtonian fluid. *Wear* 1999;232(2):213–20. [https://doi.org/10.1016/S0043-1648\(99\)00148-9](https://doi.org/10.1016/S0043-1648(99)00148-9).
- [42] Newton I. *Philosophiae naturalis principia mathematica*. G Brookman 1833.
- [43] Bair S, Winer WO. The high shear stress rheology of liquid lubricants at pressures of 2 to 200 MPa. *J Tribol* 1990;112(2):246–52. <https://doi.org/10.1115/1.2920248>.
- [44] Johnson, K. L., Tevaarwerk, J. L. Shear Behaviour of Elastohydrodynamic Oil Films. *Proc R Soc Lond A* 1977;356(1685):215–36. <https://doi.org/10.1098/rspa.1977.0129>.
- [45] Bair S, Winer WO. A rheological model for elastohydrodynamic contacts based on primary laboratory data. *J Lubr Technol* 1979;101(3):258–64. <https://doi.org/10.1115/1.3453342>.
- [46] Bair S, Winer WO. Shear strength measurements of lubricants at high pressure. *J Lubr Technol* 1979;101(3):251–7. <https://doi.org/10.1115/1.3453339>.
- [47] Carreau PJ. Rheological equations from molecular network theories. *Trans Soc Rheol* 1972;16(1):99–127. <https://doi.org/10.1122/1.549276>.
- [48] Gecim B, Winer WO. Lubricant limiting shear stress effect on EHD film thickness. *J Lubr Technol* 1980;102(2):213–20. <https://doi.org/10.1115/1.3251474>.
- [49] Eyring H. Viscosity, plasticity, and diffusion as examples of absolute reaction rates. *J Chem Phys* 1936;4(4):283–91. <https://doi.org/10.1063/1.1749836>.
- [50] Ree T, Eyring H. Theory of non-newtonian flow. I. Solid plastic system. *J Appl Phys* 1955;26(7):793–800. <https://doi.org/10.1063/1.1722098>.
- [51] Neupert T, Bartel D. Evaluation of various shear-thinning models for squalane using traction measurements, TEHD and NEMD simulations. *Lubricants* 2023;11(4):178. <https://doi.org/10.3390/lubricants11040178>.
- [52] Rodermund H. Extrapolierende Berechnung des Viskositätsverlaufes unter hohen Drücken 1980;27:3–5.
- [53] Tanner RI. *Engineering Rheology*. OUP Oxford; 2000.
- [54] Poon SY, Haines DJ. Third paper: frictional behaviour of lubricated rolling-contact elements. *Proc Inst Mech Eng* 1966;181(1):363–89. https://doi.org/10.1243/PIME_PROC_1966_181_036_02.
- [55] Czichos H. Influence of asperity contact conditions on the failure of sliding elastohydrodynamic contacts. *Wear* 1977;41(1):1–14. [https://doi.org/10.1016/0043-1648\(77\)90187-9](https://doi.org/10.1016/0043-1648(77)90187-9).
- [56] Kaneta M. Effects of surface roughness in elastohydrodynamic lubrication. *JSME Int J Ser 3 Vib Control Eng* 1992;35(4):535–46.

- [57] Bobach L, Beilicke R, Bartel D, Deters L. Thermal elastohydrodynamic simulation of involute spur gears incorporating mixed friction. *Tribol Int* 2012;48:191–206. <https://doi.org/10.1016/j.triboint.2011.11.025>.
- [58] Venner CH, Lubrecht AA. *Multilevel Methods in Lubrication*. 1st ed. Amsterdam, New York: Elsevier; 2000.
- [59] Johnson KL, Greenwood JA. Thermal analysis of an Eyring fluid in elastohydrodynamic traction. *Wear* 1980;61(2):353–74. [https://doi.org/10.1016/0043-1648\(80\)90298-7](https://doi.org/10.1016/0043-1648(80)90298-7).
- [60] Ottermann R, Steppeler T, Dencker F, Wurz MC. Degeneration effects of thin-film sensors after critical load conditions of machine components. *Machines* 2022;10(10):870. <https://doi.org/10.3390/machines10100870>.